CFD Final: Compressor Design

ME407 Computational Fluid Dynamics
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May 2026

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1 Introduction

1.1 Project Overview

Gas turbine engines use compressors to increase the pressure and density of incoming air before combustion. This allows the combustor and turbine to operate with a higher-pressure working fluid, improving the overall power output of the engine. For this project, an axial compressor was designed for a gas turbofan engine and analyzed using Computational Fluid Dynamics (CFD) in Ansys Fluent.

The purpose of the design is to meet the required compression ratio while minimizing the power required to operate the compressor. The compressor is modeled as a multi-stage rotor-stator system. The rotating blades add energy to the air, while the stationary vanes redirect the flow and help recover pressure before the next stage. CFD is used to evaluate the pressure ratio, temperature rise, and flow behavior through the compressor.

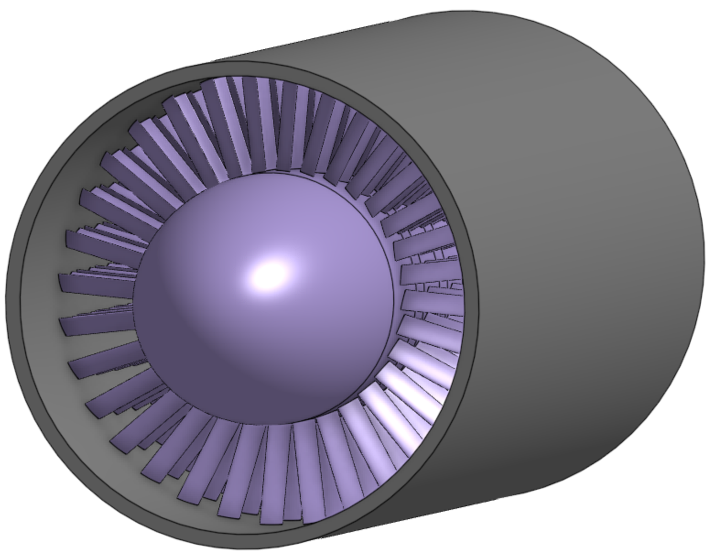


Figure 1: Overall compressor

1.2 Design Requirements & Constraints

The compressor design must satisfy the required operating conditions while remaining within the geometric and speed limits given for the project. The primary performance requirement is a minimum pressure ratio of 22:1 at the maximum design altitude of 40,000 ft. The compressor must also be evaluated at sea-level standard conditions at the same design speed.

The main design requirements are as follows:

- The compressor must provide a minimum pressure ratio of 22:1 at 40,000 ft.
- The compressor must also be evaluated at sea-level standard conditions.
- The outer diameter of the compressor must be between 5 ft and 6 ft.

- The maximum operating speed of any stage must be less than 50,000 rpm.
- The compressor may be designed as a single-stage or multi-stage unit.
- The final outlet must connect to an exhaust tube matching the exit diameter of the final stage.
- The design should minimize power required while still meeting the pressure ratio requirement.

For this design, strength of parts, material compatibility, manufacturing methods, and cost are not analyzed beyond reasonable design assumptions.

1.3 Design Overview

The selected design is an axial compressor made from 23 rotor-stator stages. Each stage consists of a rotating blade row followed by a stationary vane row. The rotor blades add work to the air, increasing the air pressure and temperature. The stator vanes then straighten the flow and reduce swirl before the air enters the next rotor stage.

The compressor has an outer diameter of 66 in, a tip clearance of 1% rotor blade length, and an overall length of 118 in. The design operating speed is 4000 rpm. The inlet hub diameter is 34 in and the exit hub diameter is 46 in, giving a gradually changing annular flow area through the compressor. The rotor blades start with an angle of attack of 46 degrees at stage 1, with the angle decreasing by one degree per stage. Similarly, the stator blades start with an angle of attack of 24 degrees at stage 1, with the angle decreasing by one degree per stage. This geometry was selected to keep the design within the required outer diameter while gradually reducing volume through each stage.

Note that Figures 2 and 1 display the 10 stage compressor used to find the number of required stages to achieve the desired total compression.

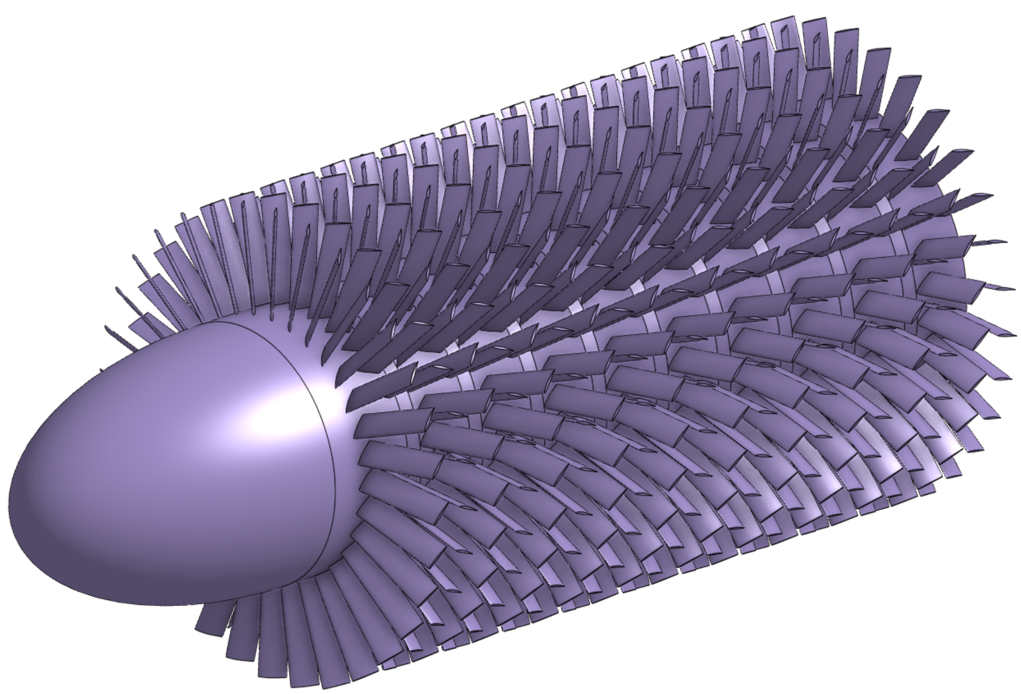


Figure 2: Isometric View of Compressor Design

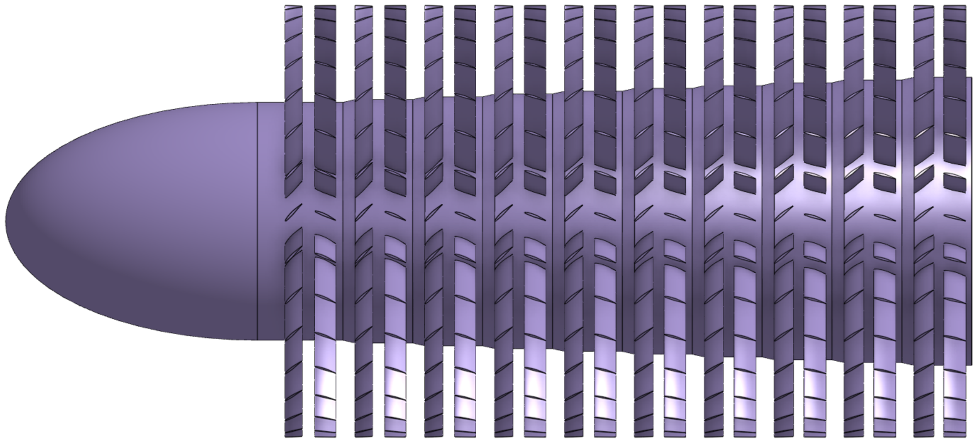


Figure 3: Side View of Compressor Design

The NACA 65-series airfoils have been proven to produce optimal results for compressor blades for inlet velocities of Mach ≤ 0.78. For the rotor and stator blade geometry, a NACA 64 410 was used. The section of each stage allocated to the rotor is 3.5 in, and the section of stage allocated to the stator is 3.25 in, with a rotor-stator axial gap of 0.75 in between. Each stage ends with a gap of 0.75 in between the end of the stator section and the end of the stage, and a gap of 1.75 in allocated to the rise between stages. The number of rotor blades per stage is 32, and the number of stator vanes per stage is 32. These values were selected to produce a practical compressor geometry while maintaining a repeatable stage layout for CFD simulation.

Table 1: Turbomachinery Design Parameters

Parameter	**Value**	**Unit**
Shell outer diameter	66	in
Stages (n)	23	–
Tip diameter (D)	62	in
Outer diameter	64	in
Compressor Length (L)	118	in
Inlet hub diameter (d_1)	34	in
Exit hub diameter (d_n)	46	in
RPM (ω_{rpm})	4000	rpm
Rotor chord	3.5	in
Stator chord	3.25	in
R-S gap (δx)	0.75	in
Rotor blades (N)	32	–
Stator vanes	32	–

2 Operating Conditions

The following air properties were assumed for the inlet conditions simulated at different altitudes.

Table 2: Properties of Air at 40,000 ft

Property	**Value**	**Unit**
Temperature	-69.7	°F
Pressure	2.72	psi
Density	5.86×10^{-4}	lbm/ft^3
Speed of Sound	968	ft/s
Heat Capacity	0.24	Btu/lbm-R
Thermal Conductivity	0.0113	Btu/h-ft-R

Table 3: Properties of Air at Sea Level

Property	**Value**	**Unit**
Temperature	59	°F
Pressure	14.7	psi
Density	2.378×10^{-3}	lbm/ft^3
Speed of Sound	1,116	ft/s
Heat Capacity	0.24	Btu/lbm-R
Thermal Conductivity	0.0146	Btu/h-ft-R

3 Hand Calculations

3.1 Shaft Geometry

The design of the compressor (of effective/tip diameter D) will involve n rotor-stator sections such that the shaft diameters (d_i) increase linearly along the axial direction through the compressor:

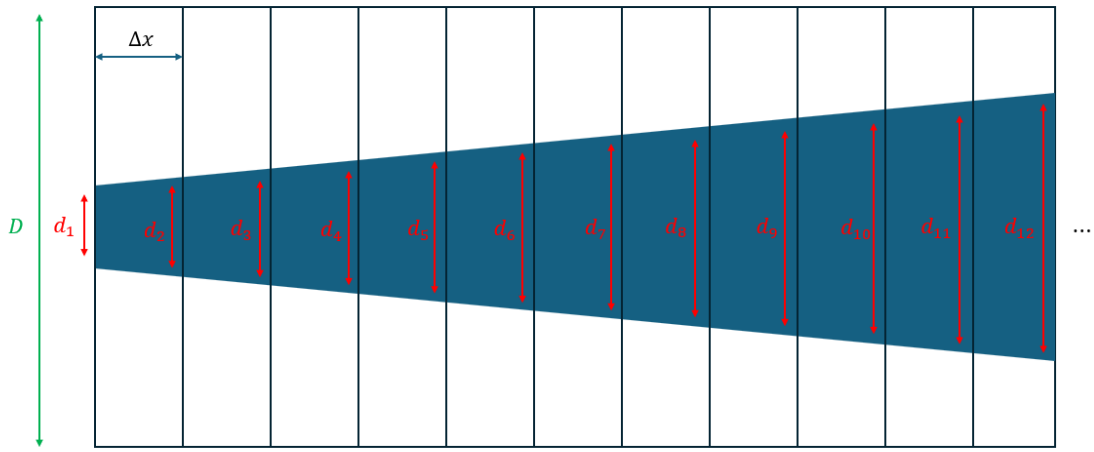


Figure 4: Shaft Slope

In particular, per the above figure, if each stage of the compressor is axially separated by a distance Δx (including the inter-stage R-S blade gap δx), then the slope of the shaft is given by:

$$S = \frac{d_{i+1} - d_i}{2\Delta x} = \text{constant}$$

Alternatively, if the compressor has a total length L, then $\Delta x = L/n$, and the shaft slope can be formulated in the following manner:

$$S = \frac{d_n - d_1}{2L} = \frac{d_n - d_1}{2n\Delta x}$$

Given an initial shaft diameter d_1, the successive diameters can be calculated according to the following equation:

$$d_i = d_1 + 2(i-1)S\Delta x$$

Moreover, the blade length ($\times 2$) in each stage can be calculated by subtracting the shaft diameter from the outer diameter of the compressor:

$$\boxed{l_{\text{blade},i} = \frac{D - d_i}{2} = \frac{D - d_1}{2} - (i-1)S\Delta x = l_{\text{blade},i-1} - S\Delta x}$$

3.2 Air Kinematics

Consider the airflow through one of the rotor-stator stages of the compressor. The geometry of this depiction will be assumed to be periodic* for simplicity as shown below:

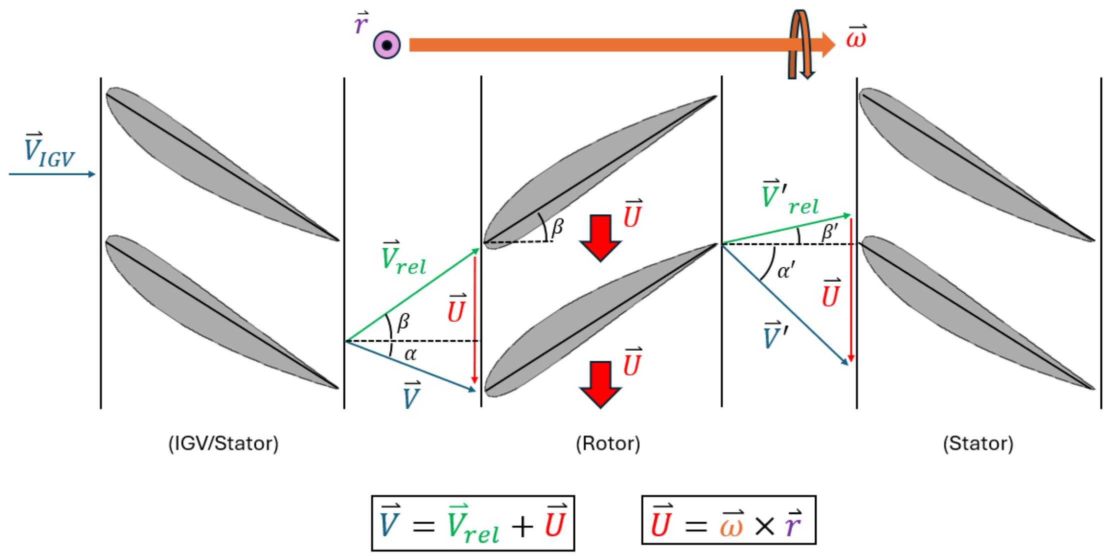


Figure 5: Velocity Diagram

In the above diagram, air flow comes in from the inlet guide vane (IGV) with some velocity $\vec{V}_{IGV}$ relative to the stator(s). The air exits with some velocity $\vec{V}$ relative to the stator right before it enters the rotor section. The rotors themselves rotate with angular velocity $\vec{\omega}$ with a lever arm $\vec{r}$, and so they acquire a linear velocity $\vec{U} = \vec{\omega} \times \vec{r}$. Note that $\omega = \frac{\pi}{30}(\omega_{\text{rpm}} \frac{\text{min}}{\text{rev}}) \frac{\text{rad}}{\text{s}}$.

The tangential direction is thus $\hat{\theta} = \vec{U}/U$, and in this regard, we may define $\vec{V}_{\text{rel}} = \vec{V} - \vec{U}$, which is the air velocity relative to the rotor. We may further decompose along the axial and tangential directions. In particular, observe from the diagram:

$$|V_\theta| = V \sin \alpha, \quad |V_{\text{rel},\theta}| = V_{\text{rel}} \sin \beta, \quad V_{\text{axial}} = V \cos \alpha, \quad \tan \beta = \frac{U - |V_{\text{rel},\theta}|}{V_{\text{axial}}} = \frac{U + V_{\text{rel},\theta}}{V_{\text{axial}}}$$

Ultimately, we assume V_{axial} to be **constant** throughout the compressor with $V_{\text{axial}} = V_0$, which is presumed to be the initial (axial) speed of the air relative to the entrance of the compressor. With all of that in mind:

$$V_\theta = V_{\text{rel},\theta} + \Delta V_\theta = V_{\text{axial}} \tan \beta$$

Once the air passes through the rotor section, it exits with a velocity $\vec{V}'$ relative to the stator(s) right before it enters the next stator section. The same decomposition can be made as follows (per the diagram):

$$V'_\theta = V'_{\text{rel},\theta} + \Delta V_\theta = V_{\text{axial}} \tan \beta'$$

Note that β roughly corresponds to the angle of attack of the rotor blade relative to the axial flow. Likewise, β' corresponds to the airfoil's trailing edge slope relative to the axial flow.

The quantity $\Delta V_\theta = V'_\theta - V_\theta$ is called the swirl velocity. Importantly, ΔV_θ is proportional to the specific work the air does on each rotor-stator stage of the compressor. To be explicit:

$$\boxed{\Delta V_\theta = |V'_\theta - V_\theta| = V_{\text{axial}} |\tan \beta' - \tan \beta| = V_0 |\tan \beta' - \tan \beta| \approx \text{constant}}$$

3.3 Compressor Dynamics

Right as air enters the compressor with ambient temperature T_0 and speed V_0, it heats up due to the initial stagnation. This stagnation temperature (T_1) acts as an upper bound for the air temperature at the inlet of the first stage of the compressor. Provided the specific heat $c_{p,\text{air}}$, assuming ideal gas conditions, it follows from energy conservation (per unit mass) that:

$$c_{p,\text{air}}T_1 + \frac{1}{2}0^2 = c_{p,\text{air}}T_0 + \frac{1}{2}V_0^2 \implies \boxed{T_1 = T_0 + \frac{V_0^2}{2c_{p,\text{air}}}}$$

Now, suppose that W_{rotor} is the work done by the air in a compressor section on its surroundings (mainly the rotor blades). Let m_{air} be the mass of the air (within that control volume), and ΔT be the temperature change across it. It directly follows from the Ideal Gas Law that (just as before):

$$W_{\text{rotor}} = m_{\text{air}}c_{p,\text{air}}\Delta T \implies w_{\text{rotor}} = W_{\text{rotor}}/m_{\text{air}} = c_{p,\text{air}}\Delta T$$

Therefore, relative to the i^{th} stage within the compressor:

$$T_{i+1} = T_i + \frac{w_{\text{rotor},i}}{c_{p,\text{air}}}$$

As mentioned in the previous subsection, the specific work that the air does within each stage is proportional to the swirl velocity. Within the i^{th} stage, given that the (rotor) blade length is $l_{\text{blade},i}$, the specific work done by the air in the i^{th} section can be given by (with a stage efficiency fudge factor η_c):

$$w_{\text{rotor},i} = \eta_c\bar{U}\Delta V_\theta = \bar{r}\omega\Delta V_\theta \approx \eta_c l_{\text{blade},i}\omega\Delta V_\theta$$

Upon synthesizing:

$$\boxed{T_{i+1} = T_i + \eta_c\frac{l_{\text{blade},i}\omega\Delta V_\theta}{c_{p,\text{air}}}} \implies \frac{T_{i+1}}{T_i} = 1 + \eta_c\frac{l_{\text{blade},i}\omega\Delta V_\theta}{c_{p,\text{air}}T_i}$$

Given that the average pressure of the i^{th} section is P_i and γ is the heat capacity ratio (adiabatic index) of the air (i.e. $\gamma = c_{p,\text{air}}/c_{v,\text{air}}$), the following thermodynamic approximation (isentropic) can be made:

$$i^{\text{th}} \text{ Compressor Pressure Ratio:} \qquad \text{CPR}_i = \frac{P_{i+1}}{P_i} \approx \left(\frac{T_{i+1}}{T_i}\right)^{\frac{\gamma}{\gamma-1}} = \left(1 + \eta_c\frac{l_{\text{blade},i}\omega\Delta V_\theta}{c_{p,\text{air}}T_i}\right)^{\frac{\gamma}{\gamma-1}}$$

The net CPR over n rotor-stator stages is given by:

$$\text{CPR}_{\text{net}} = \left(\frac{T_n}{T_1}\right)^{\frac{\gamma}{\gamma-1}} = \prod_{i=1}^{n-1}\text{CPR}_i$$

Provided that the initial pressure at the entrance of the compressor is P_1, the corresponding pressure at the exit of the compressor (after n section) is given by:

$$\boxed{P_n = \text{CPR}_{\text{net}}P_1} \implies \Delta P = P_n - P_1 = (\text{CPR}_{\text{net}} - 1)P_1$$

3.4 Compressor Efficiency

The efficiency of the compressor (η) is defined by the ratio of the total power out from the net pressure gradient ($\dot{W}_{\Delta P}$) to the power in from the working air on the rotors ($\dot{W}_{\text{rotors}}$). In particular:

$$\dot{W}_{\Delta P} = \frac{\dot{m}_{\text{air}}\Delta P}{\rho_{\text{air}}}, \qquad \dot{W}_{\text{rotors}} = N\dot{m}_{\text{air}}\sum_{i=1}^{n} w_{\text{rotor},i} = N\eta_c\dot{m}_{\text{air}}\omega\Delta V_\theta\sum_{i=1}^{n} l_{\text{blade},i}$$

And so...

$$\boxed{\eta = \frac{\dot{W}_{\Delta P}}{\dot{W}_{\text{rotors}}} = \frac{\Delta P/\Delta V_\theta}{N\eta_c\rho_{\text{air}}\omega\sum_{i=1}^{n} l_{\text{blade},i}}}$$

3.5 Preliminary Calculations & Tabulation

Table 4: Parameters Used in Preliminary Calculations (at 40,000 ft) [1]

Property	**Value**	**Units**
Heat Capacitance of Air, c_p	1005	J/kg/K
Adiabatic Index	1.4	–
Air Density @ 40,000 ft, ρ_{40k}	0.3016	kg/m^3
Air Density @ Sea Level, ρ_{sl}	1.225	kg/m^3
Number of Stages, n	23	–
Compressor Length, L	3	m
Outer Diameter, D	1.6764	m
Inlet Diameter, d_1	0.8636	m
Outlet Diameter, d_n	1.1684	m
Incoming Angle, β	46	°
Outgoing Angle, β'	35.3	°
Rotor Velocity, ω_{rpm}	4000	rpm
Assumed Stage Efficiency, η_c	1.00	–

Table 5: Initialization Inputs Used in Calculations

Property	**Metric**	**Freedom**
Initial Air Speed, V_0	267.54 m/s	878.1 ft/s
Ambient Air Temperature @ 40,000 ft, $T_{0,40k}$	216.66 K	-62.4 °F
Ambient Air Temperature @ Sea Level, $T_{0,sl}$	288.15 K	59.0 °F
Ambient Air Pressure @ 40,000 ft, $P_{1,40k}$	18750 Pa	2.72 psi
Ambient Air Pressure @ Sea Level, $P_{1,sl}$	101325 Pa	14.7 psi

Tabulated Calculations

$$S = \frac{d_n - d_1}{2L} = \frac{1.1684 \text{ m} \ - \ 0.8636 \text{ m}}{2 \ (3 \text{ m})} = 0.0508, \quad \Delta x = \frac{L}{n} = \frac{3 \text{ m}}{10} = 0.3 \text{ m}$$

Table 6: Blade Geometry

Stage Number	**Blade Length (m)**	**Blade Length (in)**
1	0.406	15.98
2	0.399	15.71
3	0.387	15.24
4	0.380	14.96
5	0.373	14.69
⋮	⋮	⋮
20	0.281	11.06
21	0.274	10.79
22	0.267	10.51
23	0.261	10.28

Upon initializing the recurrence relation involving the temperature of the various stages of the compressor (T_i), and subsequently the pressures/CPRs, with T_0 and P_1 (for both 40,000 ft and at sea level), the following values were acquired:

Table 7: Hand Calc Tabulated CPR, P, and T Values @ 40,000 ft

Stage Number	**CPR**	P (Pa)	P (psi)	T (K)	T (°F)
1	1.31	18750	2.72	252	-6
2	1.28	24500	3.55	272	30
3	1.25	31288	4.54	292	66
4	1.23	39180	5.68	311	101
5	1.21	48233	7.00	330	135
⋮	⋮	⋮	⋮	⋮	⋮
20	1.09	339471	49.22	577	579
21	1.08	368806	53.45	591	605
22	1.08	399153	57.92	604	628
23	–	430445	62.42	618	653
Net CPR	22.95	–	–	–	–
Net Power Usage	239424 hp	–	–	–	–
Net Efficiency	14.4%	–	–	–	–

Table 8: Hand Calc Tabulated CPR, P, and T Values @ Sea Level

Stage Number	**CPR**	P (Pa)	P (psi)	T (K)	T (°F)
1	1.17	101325	14.70	324	124
2	1.16	118534	17.19	339	151
3	1.15	137407	19.93	353	176
4	1.14	157969	22.91	368	203
5	1.13	180240	26.14	382	228
⋮	⋮	⋮	⋮	⋮	⋮
20	1.06	708576	102.76	564	556
21	1.06	754621	109.42	575	575
22	1.06	801602	116.27	585	593
23	–	849425	123.20	594	610
Net CPR	8.38	–	–	–	–
Net Power Usage	43506 hp	–	–	–	–
Net Efficiency	35.2%	–	–	–	–

With some hindsight, the stage efficiency η_c could have been preemptively tuned to match the CFD analysis. Specifically, it could have been increased (beyond the assumed 100%). This is a byproduct of the loose assumption that the swirl velocity ΔV_θ remained constant throughout the compressor as provided. This is all to say (again), η_c is just a fudge factor.

From the hand calculations, the air compressor will have a higher CPR at 40,000 ft than it will at sea level. However, the compressor becomes much less efficient at the higher altitude (roughly by a factor of 2.44). Although counterintuitive, keeping the rotors spinning at the same angular speed requires more bang for the same buck the higher up one goes in altitude.

3.6 CPR Interpolation Model

Given the recursive relationship between T_{i+1} and T_i, upon inspection, because $l_{\text{blade},i}$ shrinks linearly (but slowly) in i, we can make the approximation:

$$T_i \approx T_1 + bT_1(i-1) \quad \text{for some} \quad b \in \mathbb{R}$$

Therefore, we can further deduct:

$$\text{CPR}_i \approx \left(1 + \frac{(\text{constant})}{T_i}\right)^{\frac{\gamma}{\gamma-1}} = \left(1 + \frac{\frac{\gamma-1}{\gamma} r T_1}{T_1 + bT_1(i-1)}\right)^{\frac{\gamma}{\gamma-1}} \approx 1 + \frac{r}{1+b(i-1)} \quad \text{for some} \quad r \in \mathbb{R}$$

We can thus apply a regression model of the following form to fit the pressure data as needed:

$$\boxed{\text{CPR}_i \sim 1 + \frac{r}{1+b(i-1)}}$$

In order to validate this regression, it will be fitted to the (CPR) hand calculations that were tabulated in the previous section (@ 40,000 ft). If the R^2 value is ≈ 1, and the theory behind the hand calculations is valid, then we can apply the regression to the CFD model. The hand calc results are as such:

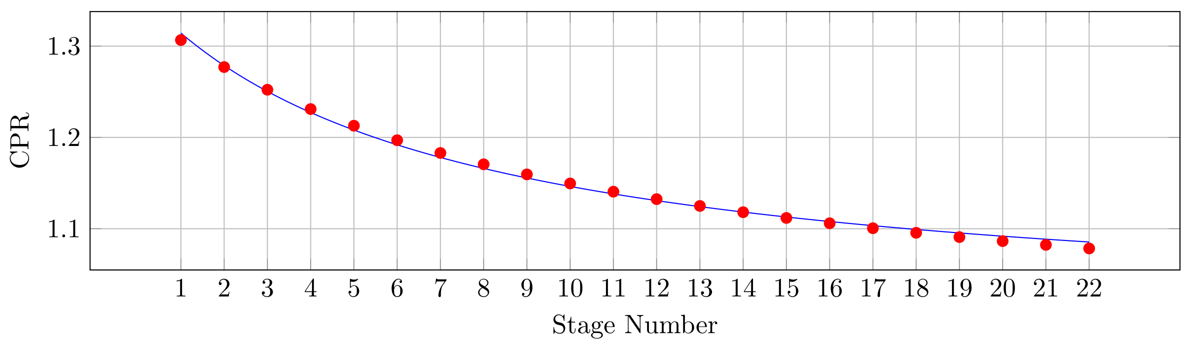


Figure 6: Hand Calc Regression @ 40,000 ft ($R^2 = 0.9957$)

Obviously, $R^2 = 0.9957 \approx 1$, which implies we can apply the regression in the appropriate context. Because the R^2 value is so close to unity, we need not interpolate through more than two data points; two points are sufficient to determine r and b.

Assuming this *power* law to be valid, knowing how the compressor operates at the first two stages can determine its performance down the line. In particular:

$$\text{CPR}_1 \sim 1 + r \implies r \approx \text{CPR}_1 - 1$$

And so...

$$\text{CPR}_2 \sim 1 + \frac{r}{1+b} \implies b \approx \frac{r}{\text{CPR}_2 - 1} - 1 = \frac{\text{CPR}_1 - 1}{\text{CPR}_2 - 1} - 1$$

Overall, upon synthesis & rearrangement, this suggests:

$$\boxed{\text{CPR}_i \approx 1 + \frac{\text{CPR}_1 - 1}{1 + \left(\frac{\text{CPR}_1 - 1}{\text{CPR}_2 - 1} - 1\right)(i-1)}}$$

This interpolation will be used to model the stages superseding the initial two.

Moreover, we could even squeeze out the (predicted) second compressor pressure ratio using the first if absolutely needed. In particular, observe from the hand calculation model:

$$\eta_c \frac{l_{\text{blade},1}\omega\Delta V_\theta}{c_{p,\text{air}}T_1} = \text{CPR}^{\frac{\gamma-1}{\gamma}} - 1, \quad \text{and} \quad \text{CPR}_2 = \left(1 + \eta_c \frac{l_{\text{blade},2}\omega\Delta V_\theta}{c_{p,\text{air}}T_2}\right)^{\frac{\gamma}{\gamma-1}}$$

Using the recurrence relations for T_i and $l_{\text{blade},i}$:

$$\eta_c \frac{l_{\text{blade},2}\omega\Delta V_\theta}{c_{p,\text{air}}T_2} = \frac{\eta_c(l_{\text{blade},1} - S\Delta x)\omega\Delta V_\theta}{c_{p,\text{air}}T_1 + \eta_c l_{\text{blade},1}\omega\Delta V_\theta} \approx \frac{\eta_c l_{\text{blade},1}\omega\Delta V_\theta}{c_{p,\text{air}}T_1}\left(1 - \frac{S\Delta x}{l_{\text{blade},1}}\right)\left(1 - \frac{\eta_c l_{\text{blade},1}\omega\Delta V_\theta}{c_{p,\text{air}}T_1}\right)$$

After substituting and simplifying, it follows that:

$$\boxed{\text{CPR}_2 \approx \left(1 + \left(\text{CPR}_1^{\frac{\gamma-1}{\gamma}} - 1\right)\left(1 - \frac{S\Delta x}{l_{\text{blade},1}}\right)\left(2 - \text{CPR}_1^{\frac{\gamma-1}{\gamma}}\right)\right)^{\frac{\gamma}{\gamma-1}}}$$

This is a method of computing the second compressor compression ratio from the first (in conjunction with the compressor geometry & thermodynamic properties of air). Once the second CPR value is obtained, the rest of the CPR values can be interpolated through via the two-point interpolation formula already derived.

How can we be sure this (above) interpolation works? Let's compare its output to the result of the CFD calculation in advance (note that the CFD results will be analyzed separately later). In the compressor simulation at 40,000 ft, we concluded that $\text{CPR}_1 = 1.223$ and $\text{CPR}_2 = 1.212$. Plugging in $\text{CPR}_1 = 1.223$, γ, S, and Δx into the above interpolation results in $\text{CPR}_2 \approx 1.205$. In this regard, the interpolation underestimates CPR_2 by a factor of (roughly) 1.0056. This is quite close; to the point where we can use it to predict future values when working at sea level.

4 Analysis Methodology

4.1 Overview

The compressor was analyzed using Ansys TurboGrid and Ansys Fluent Turbo Workflow. Because the full compressor contains multiple rotor-stator stages, each stage was first modeled individually. This reduced the complexity of the simulation and allowed the pressure ratio of each stage to be found before extrapolating the total compressor performance.

The first rotor and stator were meshed separately in TurboGrid. These meshes were then imported into Fluent Turbo Workflow and connected as a single compressor stage. The stage was first simulated at low rotational speed and low outlet pressure to produce a stable initial solution. After convergence was reached, the rotational speed was increased in steps using the previous step as initialization until the design operating speed was reached. Once the operating speed was reached, the outlet pressure was increased incrementally. The maximum stable outlet pressure before stall was used to determine the maximum pressure ratio of the stage.

The pressure ratio from stage 1 was calculated using the inlet and outlet conditions from the converged Fluent solution. The outlet profile from stage 1 was then exported and used as the inlet condition for stage 2. This allowed the second stage to be simulated using the actual flow properties leaving the first stage instead of idealized freestream air properties.

The same process was then repeated for stage 2. The second rotor and stator were meshed separately in TurboGrid, imported into Fluent Turbo Workflow, and simulated as a rotor-stator pair. The inlet condition was set using the exported outlet profile from stage 1. The stage 2 simulation was again ramped from low speed to the design rotational speed, then the outlet pressure was increased until the maximum stable pressure ratio was found.

After obtaining the pressure ratio from the simulated stages, the hand calculation model was used to estimate the total compressor pressure ratio after n stages. This allowed the required number of stages to be determined without simulating every stage individually. The CFD simulations therefore provided stage-level performance data, while the hand calculations were used to extrapolate the total compressor performance.

4.2 Simulation Procedure

The simulation process was performed in the following order:

1. Mesh rotor 1 and stator 1 separately in TurboGrid.
2. Import both meshes into Fluent Turbo Workflow.
3. Simulate stage 1 at low rpm and low outlet pressure.
4. Increase the rotational speed in steps until the design operating rpm is reached.
5. Increase the outlet pressure in steps until the highest stable pressure ratio is reached before stall.
6. Record the stage 1 pressure ratio and export the stage 1 outlet profile.
7. Mesh rotor 2 and stator 2 separately in TurboGrid.
8. Import both meshes into Fluent Turbo Workflow.
9. Use the exported stage 1 outlet profile as the stage 2 inlet condition.
10. Repeat the rpm ramping and outlet pressure ramping process for stage 2.
11. Use the simulated stage pressure ratios with the hand calculation model to estimate the number of stages required for the full compressor.

This method was selected because compressor stages are highly sensitive to inlet flow conditions. Using the outlet profile from the first stage as the inlet for the second stage gives a more realistic estimate of downstream stage behavior than applying raw atmospheric inlet conditions to every stage.

4.3 Boundary Conditions

The inlet to the first stage was modeled using air properties corresponding to the selected altitude condition. The inlet flow was assumed to be subsonic, as allowed by the project assumptions. The outlet was modeled as a pressure outlet. During the simulation process, this outlet pressure was increased until the stage could no longer produce a stable solution without stalling.

The rotor domain was assigned a rotational speed of 4000 rpm. The stator domain was stationary. The rotor-stator interface was modeled using the Turbo Workflow interface settings in Fluent. The walls of the blades, hub, and casing were modeled as no-slip walls.

For stage 2, the inlet was not set directly from standard atmospheric properties. Instead, the exported outlet profile from stage 1 was used as the stage 2 inlet. This profile included the flow behavior leaving the first stage, allowing the second stage simulation to account for the pressure, velocity, temperature, and swirl produced by stage 1.

Table 9: Primary CFD Boundary Conditions

Boundary	Condition
Stage 1 inlet	altitude air properties
Stage 1 outlet	Pressure outlet
Stage 1 rotor	Rotating domain, 4000 rpm
Stage 1 stator	Stationary domain
Stage 2 inlet	Exported stage 1 outlet profile
Stage 2 outlet	Pressure outlet
Stage 2 rotor	Rotating domain, 4000 rpm
Stage 2 stator	Stationary domain
Blade, hub, and casing surfaces	No-slip walls

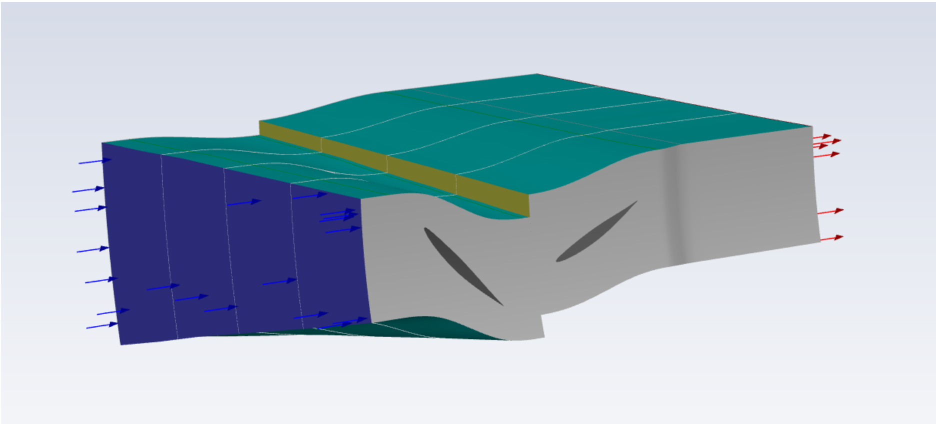


Figure 7: Boundary Conditions on Stage 1

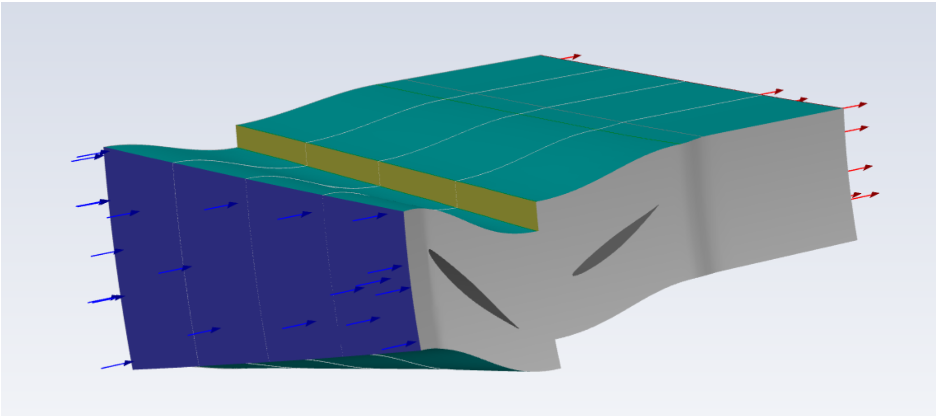


Figure 8: Boundary Conditions on Stage 2

4.4 Convergence Criteria

Each simulation was monitored using residuals and compressor performance monitors. A solution was considered usable once the residuals reached stable values and the pressure ratio monitor stopped changing significantly with additional iterations. During the rpm ramping process, each new case was initialized from the previous lower-speed solution. This helped reduce convergence time and prevented the solver from starting from an unrealistic flow field.

After the design rpm was reached, the outlet pressure was increased in small increments. If the pressure ratio remained stable and the flow field did not show signs of stall, the outlet pressure was increased again. When the simulation became unstable or showed stall behavior, the previous converged case was taken as the maximum operating pressure ratio for that stage.

5 Mesh

5.1 Overview

The rotor and stator passages were meshed separately in TurboGrid. This approach was used because the rotor and stator have different blade geometries and different flow domains. Each blade row was treated as its own turbomachinery passage before being imported into Fluent Turbo Workflow.

The mesh quality was checked for each component before simulation, according to TurboGrid mesh quality metrics. The quantities reviewed were orthogonality angle, minimum face angle, maximum face angle, edge length ratio, element volume ratio, minimum volume, connectivity number, and skewness. The mesh was considered acceptable if the overall mesh statistics were reasonable for a turbomachinery simulation. All meshes were kept under 750 thousand elements to reduce simulation time.

5.2 Stage 1 Rotor Mesh

The stage 1 rotor was meshed in TurboGrid using the rotor blade geometry and hub/shroud boundaries. The mesh was refined near the blade leading edge, trailing edge, pressure side, suction side, hub, and shroud. This refinement was used to better capture the high velocity gradients and pressure changes around the rotating blade.

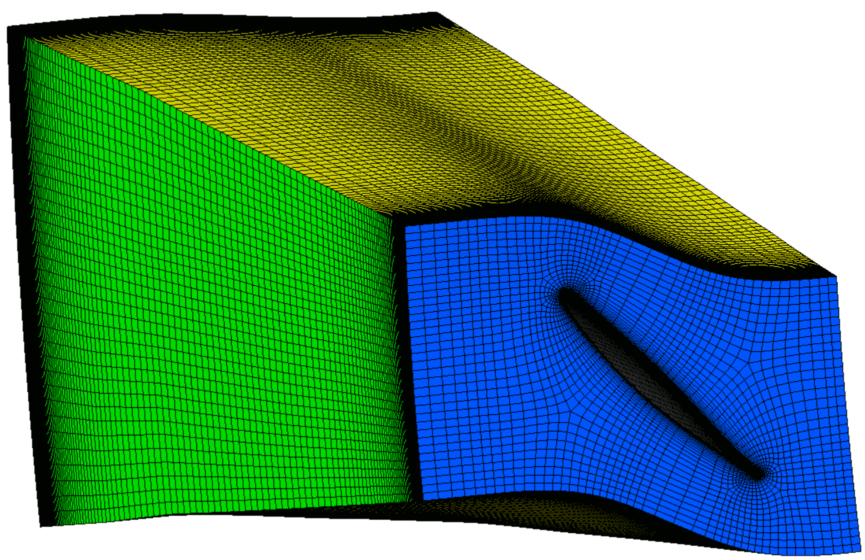


Figure 9: Stage 1 Rotor Mesh

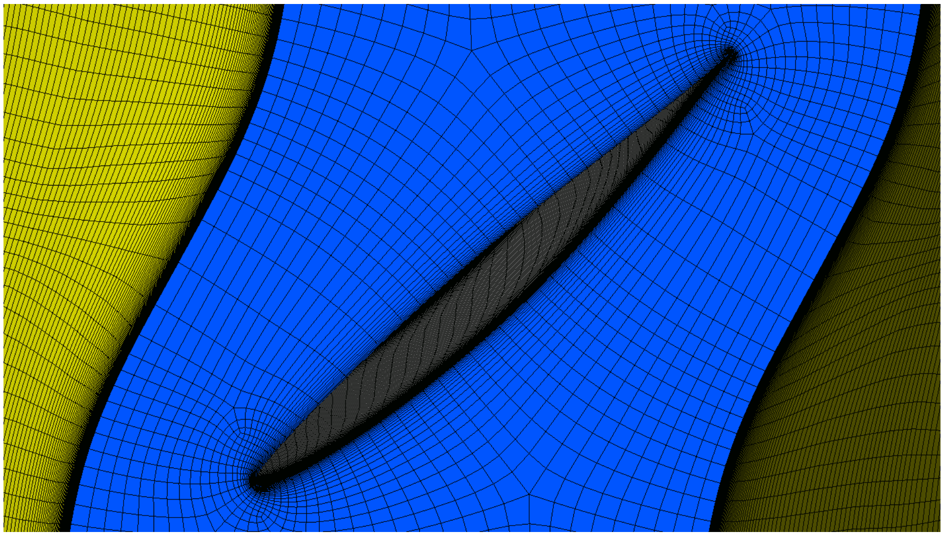
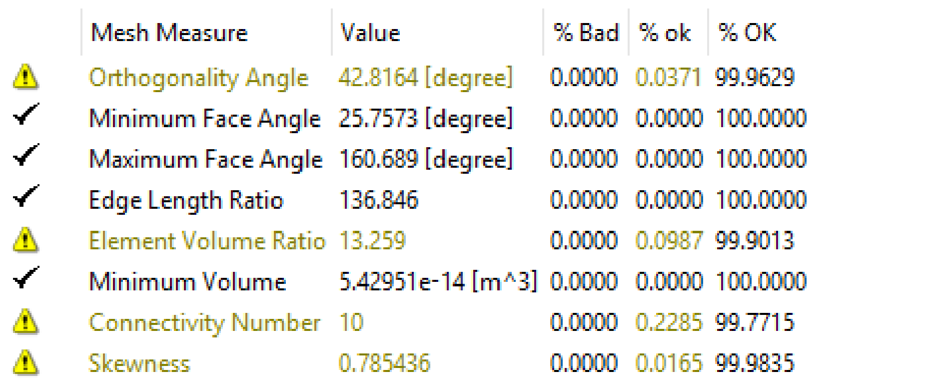


Figure 10: Stage 1 Rotor Mesh Detail



	Mesh Measure	Value	% Bad	% ok	% OK
⚠	Orthogonality Angle	42.8164 [degree]	0.0000	0.0371	99.9629
✓	Minimum Face Angle	25.7573 [degree]	0.0000	0.0000	100.0000
✓	Maximum Face Angle	160.689 [degree]	0.0000	0.0000	100.0000
✓	Edge Length Ratio	136.846	0.0000	0.0000	100.0000
⚠	Element Volume Ratio	13.259	0.0000	0.0987	99.9013
✓	Minimum Volume	5.42951e-14 [m^3]	0.0000	0.0000	100.0000
⚠	Connectivity Number	10	0.0000	0.2285	99.7715
⚠	Skewness	0.785436	0.0000	0.0165	99.9835

Figure 11: Stage 1 Rotor Mesh Quality

The mesh quality results showed that at least 99.7% of elements were acceptable. Based on these values, the mesh was considered acceptable for the stage 1 simulation.

5.3 Stage 1 Stator Mesh

The stage 1 stator was meshed separately from the rotor using the stator vane geometry. Since the stator is stationary, the mesh was designed to capture flow turning and pressure recovery after the rotor.

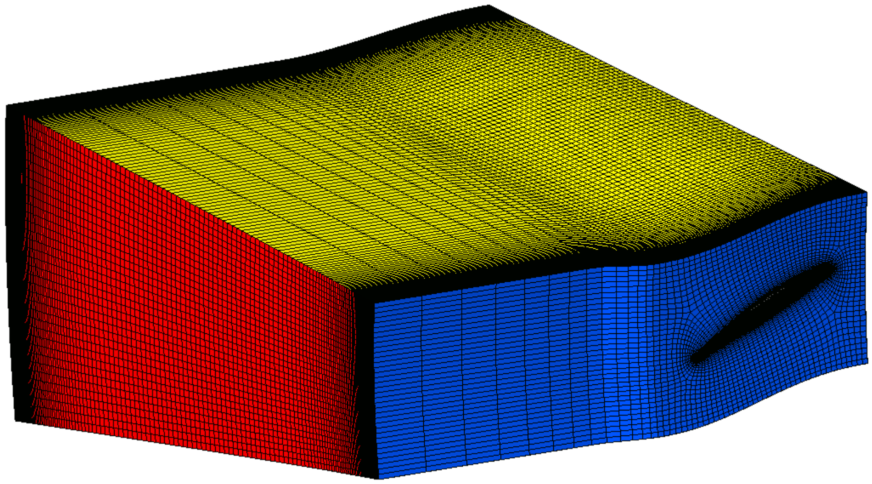


Figure 12: Stage 1 Stator Mesh

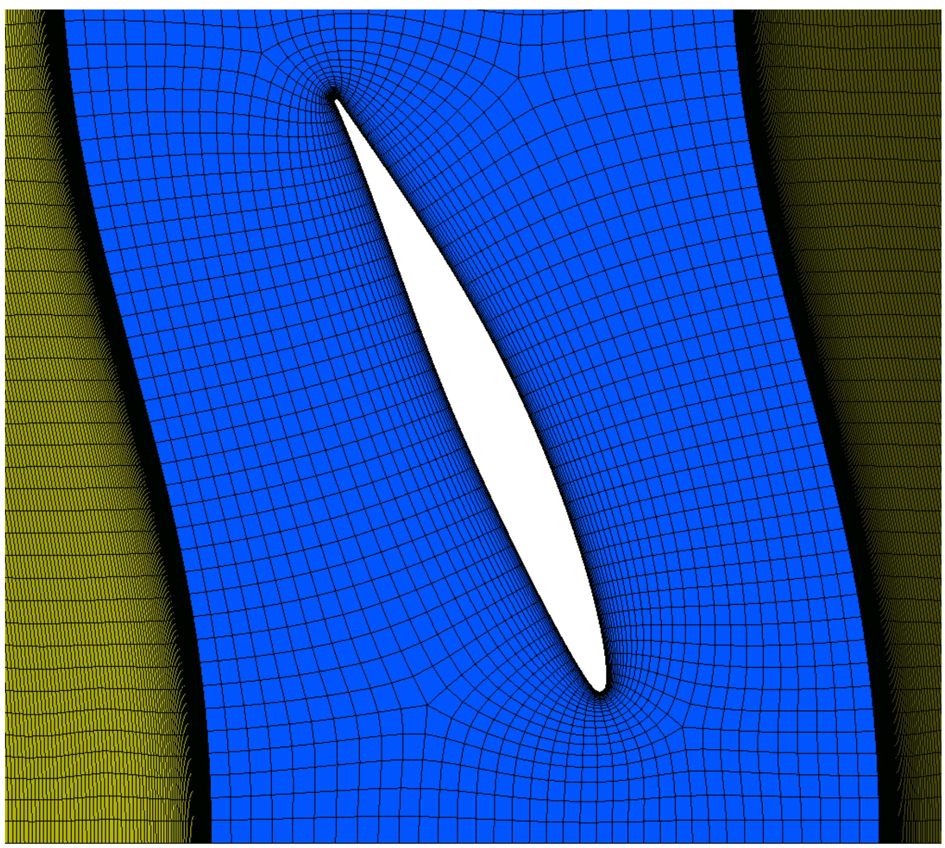


Figure 13: Stage 1 Stator Mesh Detail

Mesh Measure	Value	% Bad	% ok	% OK
Orthogonality Angle	41.6489 [degree]	0.0000	0.0134	99.9866
Minimum Face Angle	18.788 [degree]	0.0000	0.0000	100.0000
Maximum Face Angle	179.574 [degree]	0.0069	0.0078	99.9853
Edge Length Ratio	304.198	0.0000	0.0000	100.0000
Element Volume Ratio	11.5703	0.0000	0.0691	99.9309
Minimum Volume	9.46785e-13 [m^3]	0.0000	0.0000	100.0000
Connectivity Number	10	0.0000	0.1114	99.8886
Skewness	1.02721	0.0291	0.0378	99.9331

Figure 14: Stage 1 Stator Mesh Quality

The mesh quality results showed that at least 99.88% of elements were acceptable. Based on these values, the mesh was considered acceptable for the stage 1 simulation.

5.4 Stage 2 Rotor Mesh

The stage 2 rotor was meshed using the same overall TurboGrid process as the stage 1 rotor.

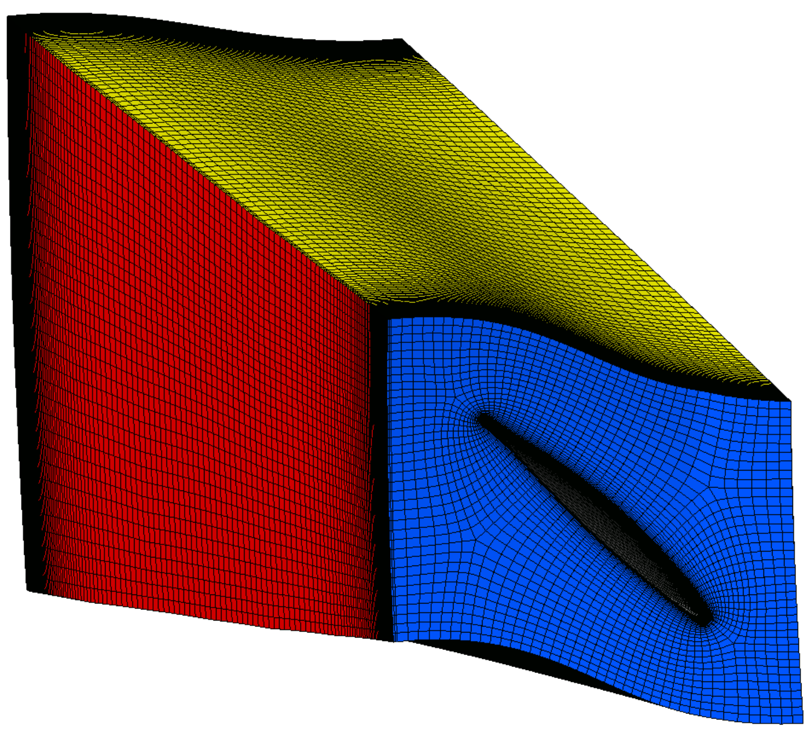


Figure 15: Stage 2 Rotor Mesh

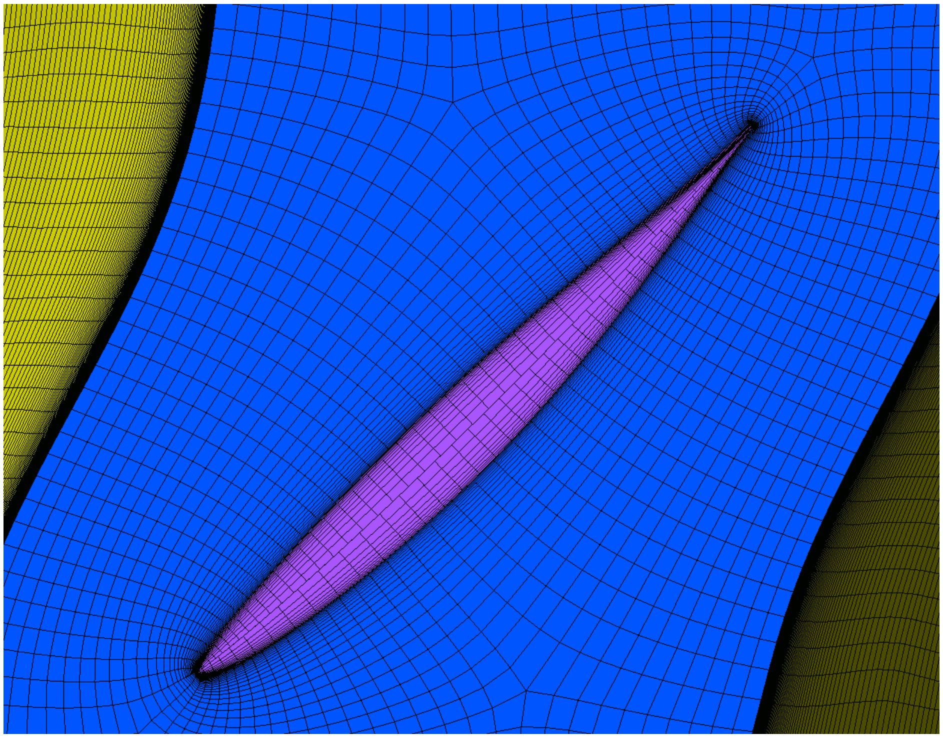


Figure 16: Stage 2 Rotor Mesh Detail

	Mesh Measure	Value	% Bad	% ok	% OK
!	Orthogonality Angle	3.67241 [degree]	0.0507	0.2045	99.7448
⚠	Minimum Face Angle	7.60791 [degree]	0.0000	0.0346	99.9654
!	Maximum Face Angle	179.23 [degree]	0.0024	0.0387	99.9589
✓	Edge Length Ratio	149.533	0.0000	0.0000	100.0000
!	Element Volume Ratio	54.6876	0.0104	0.1521	99.8375
✓	Minimum Volume	3.17431e-14 [m^3]	0.0000	0.0000	100.0000
⚠	Connectivity Number	10	0.0000	0.1182	99.8818
!	Skewness	2.99482	0.0541	0.1077	99.8382

Figure 17: Stage 2 Rotor Mesh Quality

The mesh quality results showed that at least 99.7% of elements were acceptable. Based on these values, the mesh was considered acceptable for the stage 2 simulation.

5.5 Stage 2 Stator Mesh

The stage 2 stator was meshed separately to capture the pressure recovery and flow redirection after the second rotor.

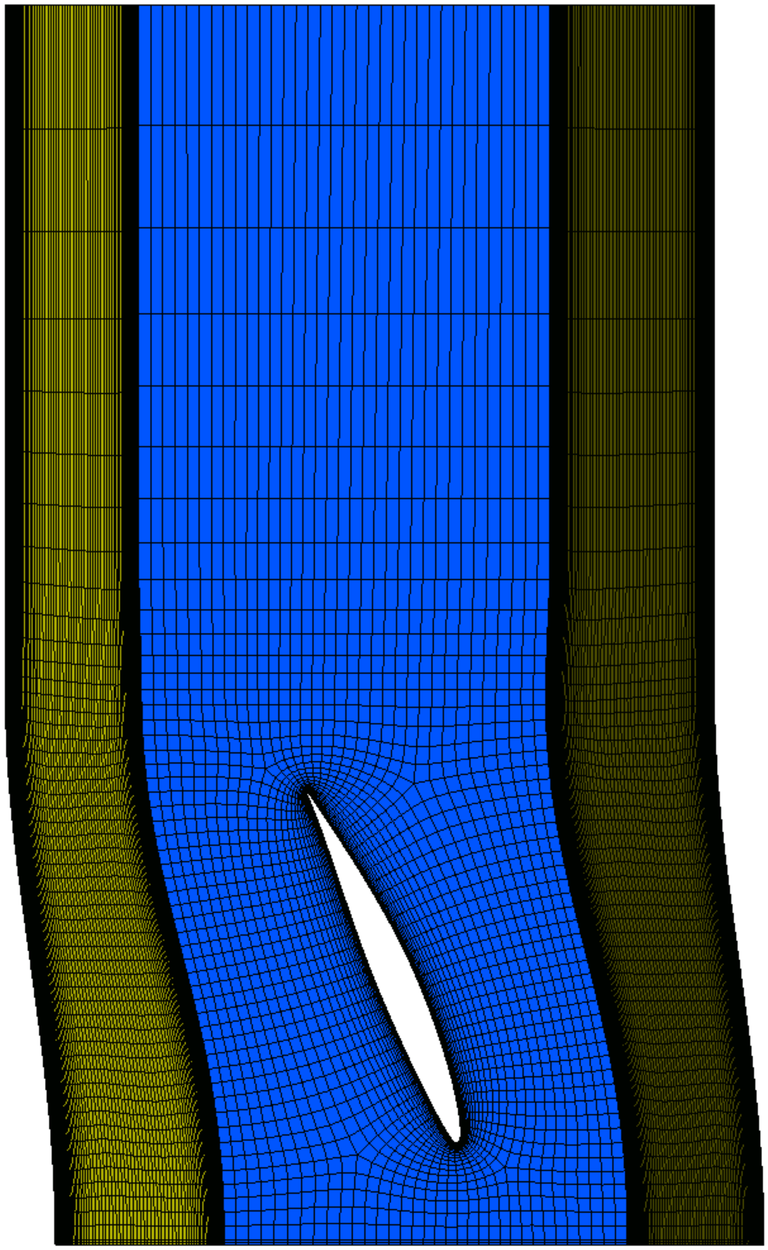


Figure 18: Stage 2 Stator Mesh

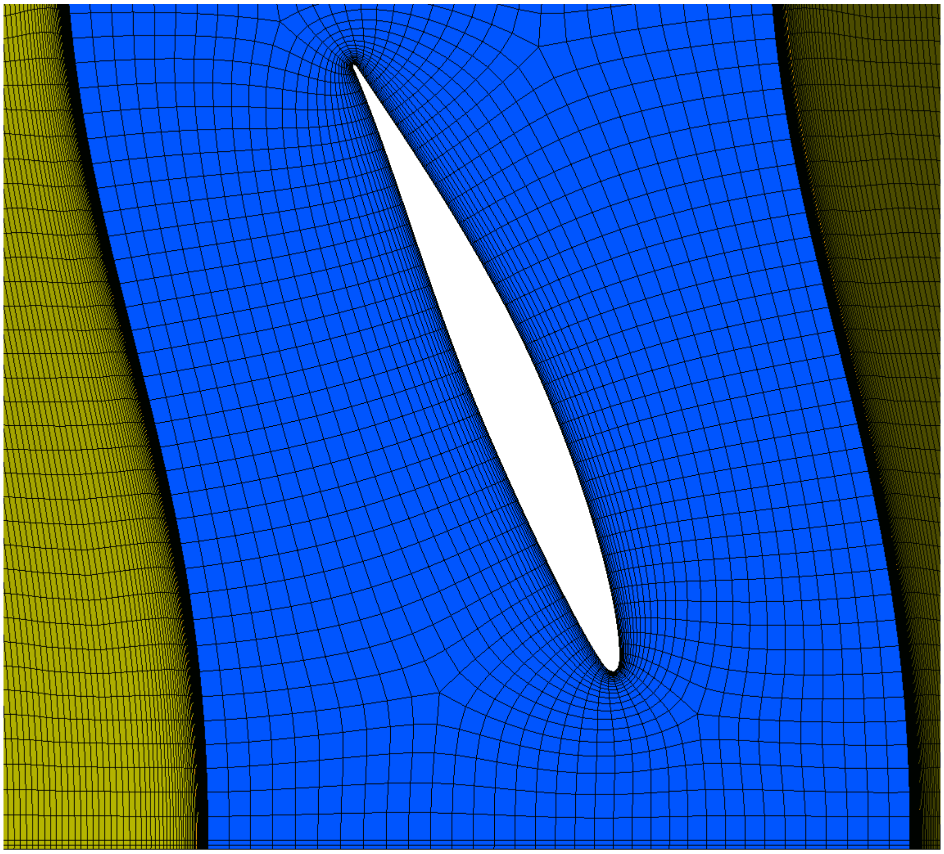


Figure 19: Stage 2 Stator Mesh Detail

	Mesh Measure	Value	% Bad	% ok	% OK
✓	Orthogonality Angle	50.5315 [degree]	0.0000	0.0000	100.0000
✓	Minimum Face Angle	43.9214 [degree]	0.0000	0.0000	100.0000
✓	Maximum Face Angle	139.832 [degree]	0.0000	0.0000	100.0000
✓	Edge Length Ratio	232.97	0.0000	0.0000	100.0000
✓	Element Volume Ratio	2.75627	0.0000	0.0000	100.0000
✓	Minimum Volume	2.00186e-12 [m^3]	0.0000	0.0000	100.0000
⚠	Connectivity Number	10	0.0000	0.1175	99.8825
✓	Skewness	0.553693	0.0000	0.0000	100.0000

Figure 20: Stage 2 Stator Mesh Quality

The mesh quality results showed that at least 99.88% of elements were acceptable. Based on these values, the mesh was considered acceptable for the stage 2 simulation.

6 Simulation Convergence

6.1 40,000 ft

Simulation convergence was evaluated using residual plots, and monitors, primarily pressure ratio monitors and mass flow rate monitors. The residuals were used to confirm that the governing equations reached stable values. The pressure ratio monitor was used to determine whether the compressor stage reached a steady operating point. The mass flow rate monitor was used to check that the flow rates were stable for a simulation.

For each stage, the solution was first stabilized at a lower rotational speed and lower outlet pressure. The rotational speed was then increased until the operating speed was reached. After this, the outlet pressure was increased until the maximum stable pressure ratio was found. The final converged case before stall was used as the reported stage pressure ratio.

6.1.1 Stage 1 Convergence

The residuals for the final stage 1 case are shown in Figure 21, and the monitors in Figures 46 and 23. Other monitors can be found in the Appendix. The iterative process of simulations are reflected in the residual and monitors, with residuals and monitors stabilizing between each step of the simulation.

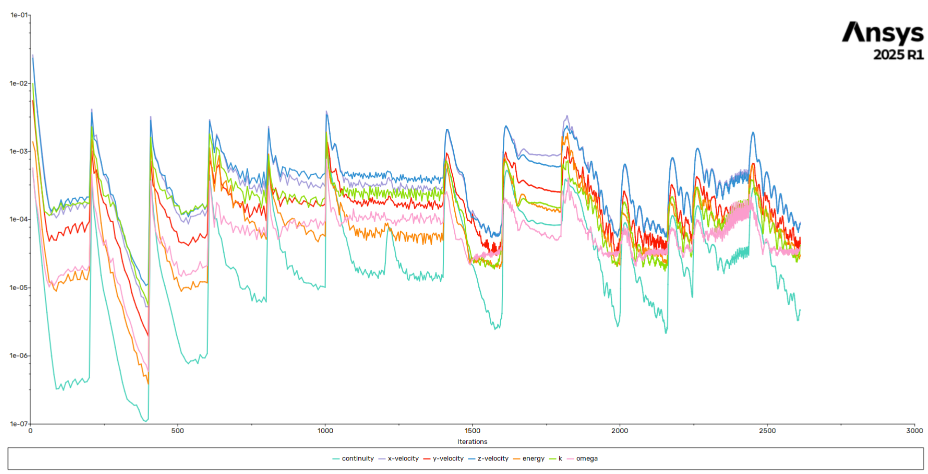


Figure 21: Stage 1 Residuals

By inspection, the compressor was pushed until it stalled, and the residuals fell below the convergence criteria for our compressor model.

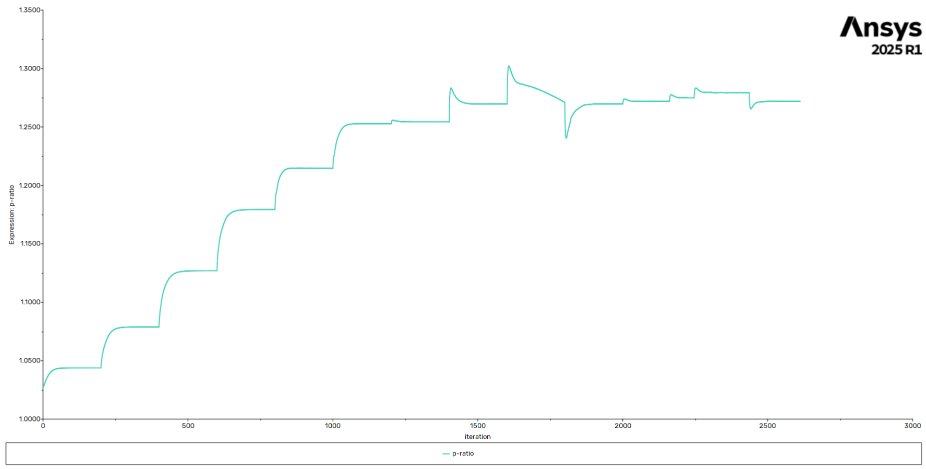


Figure 22: Stage 1 Pressure Ratio Monitor

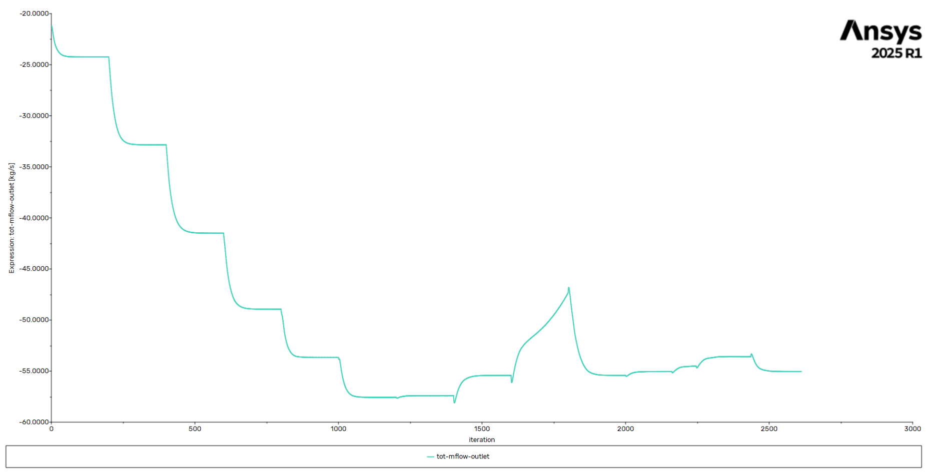


Figure 23: Stage 1 Mass Flow Rate Monitor

6.1.2 Stage 2 Convergence

The stage 2 simulation used the exported outlet profile from stage 1 as its inlet condition. This allowed the second stage to account for the pressure, temperature, velocity, and swirl leaving the first stage. The same convergence process was used for stage 2.

The residuals for stage 2 are shown in Figure 24. The pressure ratio monitor for stage 2 is shown in Figure 25. The mass flow rate monitor for stage 2 is shown in Figure 26. Other monitors can be found in the appendix. Again, the iterative process can be seen in the residuals, and the monitors were stabilized before each step was taken.

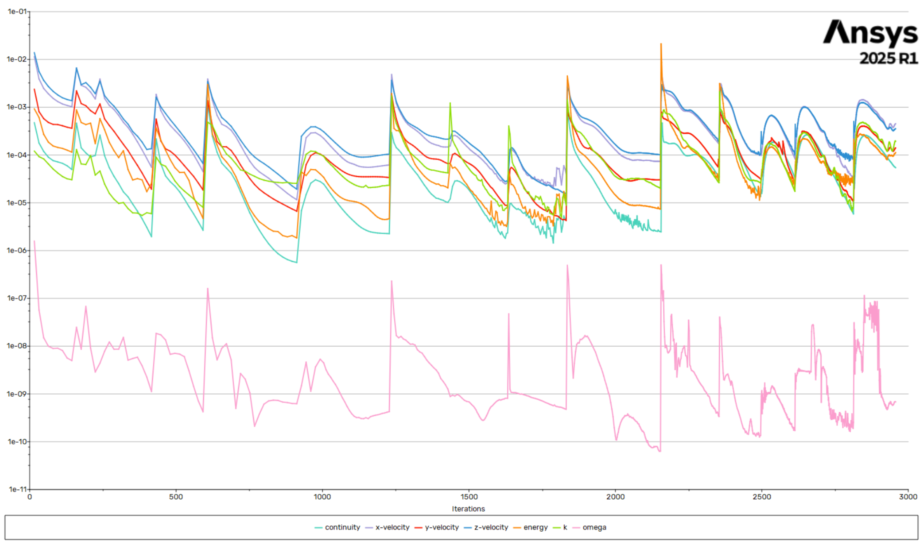


Figure 24: Stage 2 Residuals

Once again, the compressor was pushed until it stalled, and the residuals fell below the convergence criteria for our compressor model.

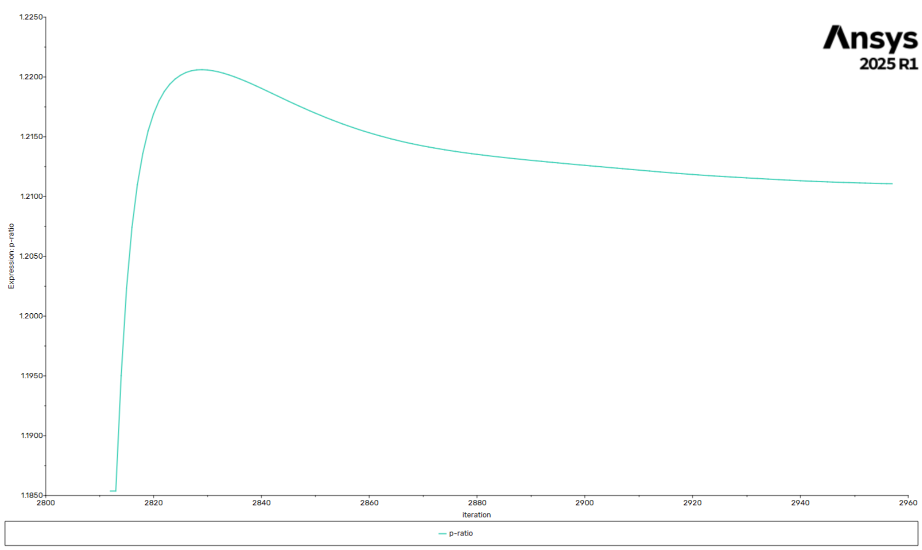


Figure 25: Stage 2 Pressure Ratio Monitor

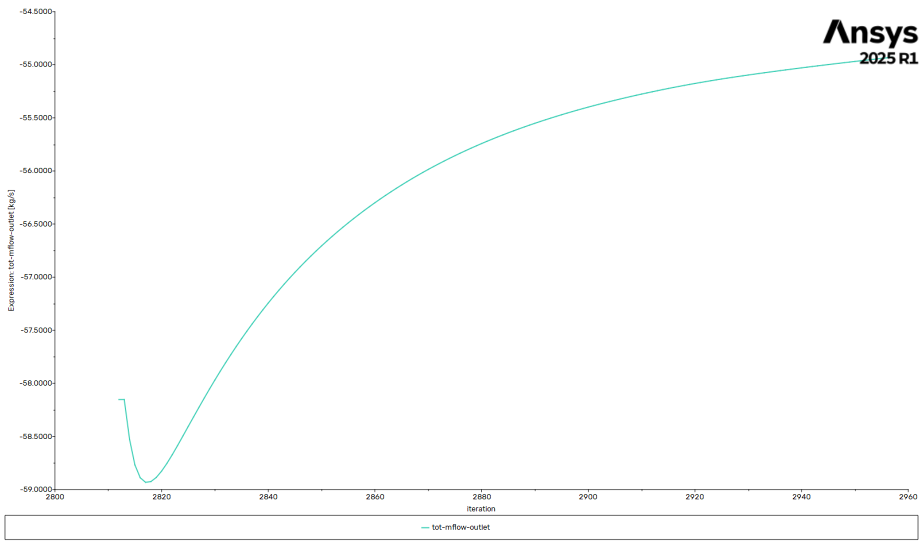


Figure 26: Stage 2 Mass Flow Rate Monitor

6.2 Sea Level

6.2.1 Stage 1 Convergence

The compressor was simulated at sea level by the same covergence criteria. The same process and convergence can be seen in the residuals and monitors below.

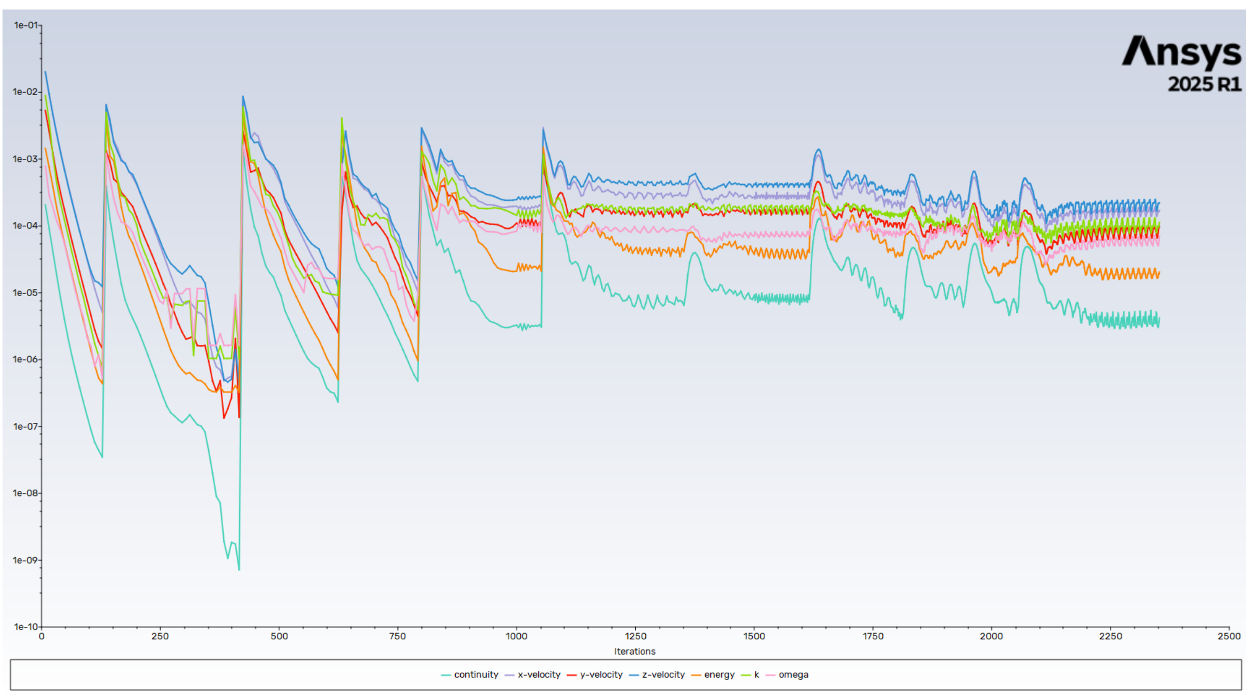


Figure 27: Stage 1 Residuals

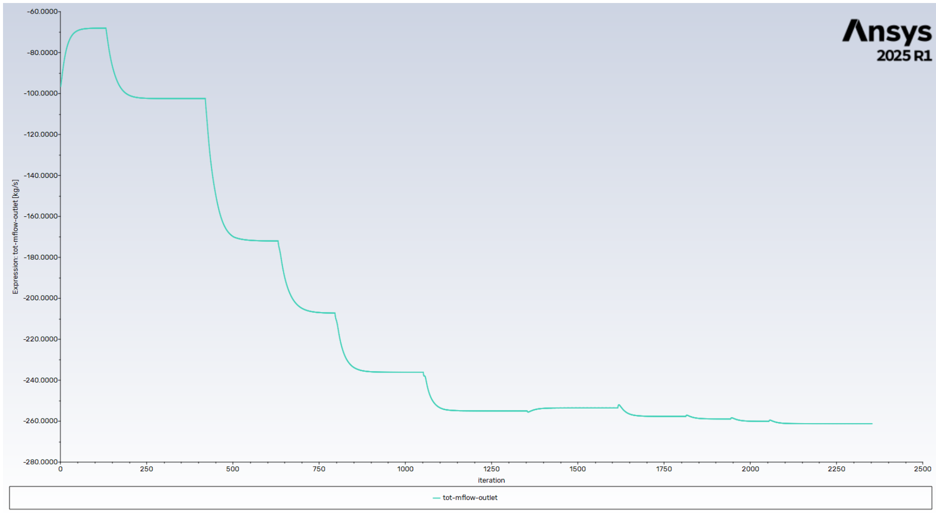


Figure 28: Stage 1 Mass Flow Rate Monitor

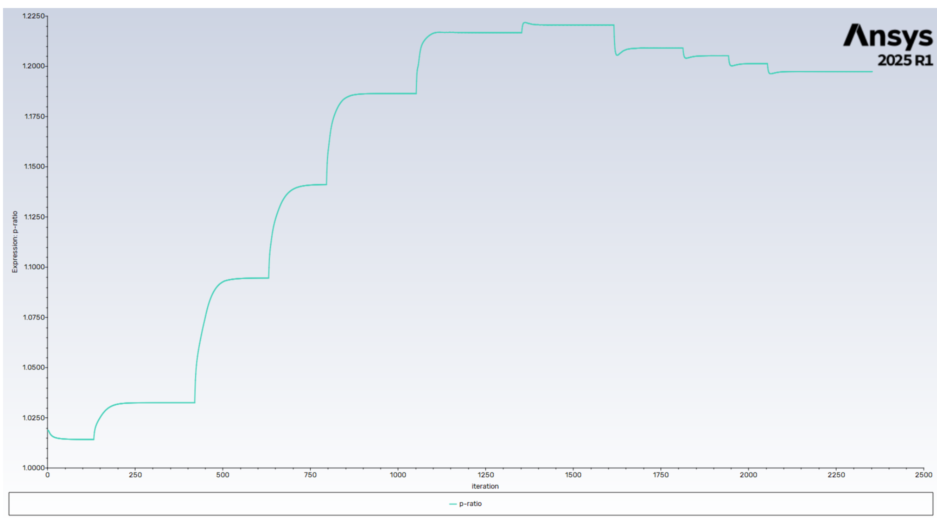


Figure 29: Stage 1 Compressor Pressure Ratio

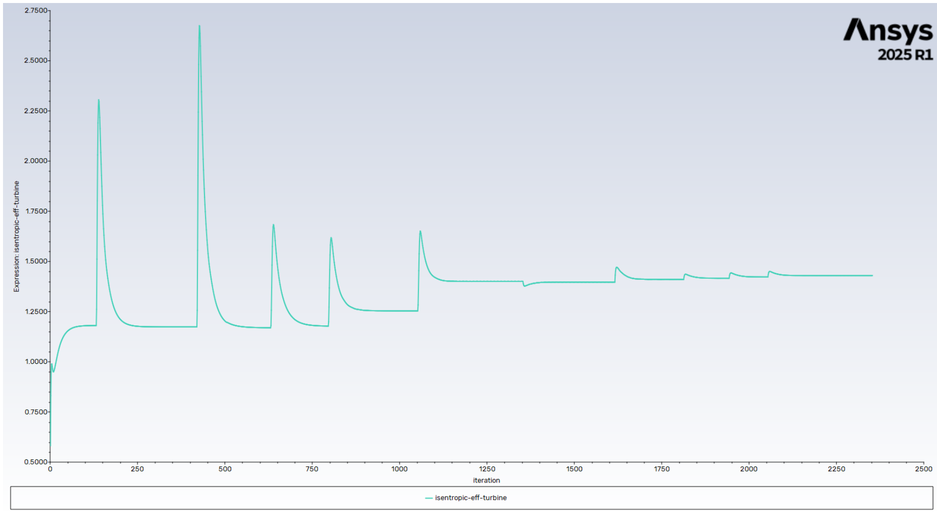


Figure 30: Stage 1 Isentropic Efficiency

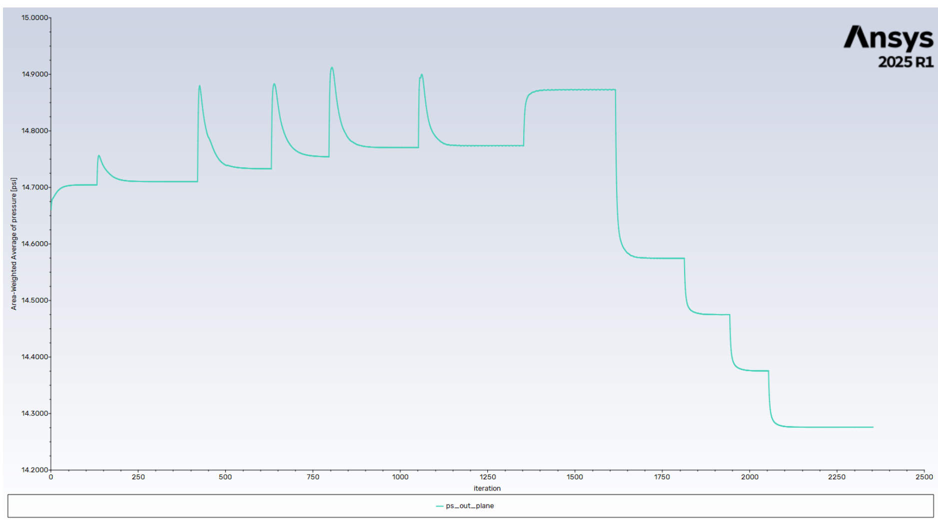


Figure 31: Stage 1 Pressure Outlet

7 Simulation Results

7.1 Results Summary

Table 10: Tabulated Results – Temperature, CPR, and Operating Efficiency

Parameter	Value		
	Stage 1 at 40,000 ft	Stage 2 at 40,000 ft	Stage 1 at Sea Level
Entrance temperature [°F]	-69.7	80.33	59
Exit temperature [°F]	-31.17	117.6	98.1
Entrance Pressure [psi]	2.72	3.44	14.7
Exit Pressure [psi]	3.32	4.17	17.6
Pressure Increase [psi]	0.604	0.73	2.9
Compressor Pressure Ratio (CPR)	1.22	1.21	1.19
Isentropic operating efficiency (%)	1.67	0.81	1.43

7.2 At 40,000 feet

7.2.1 Stage 1

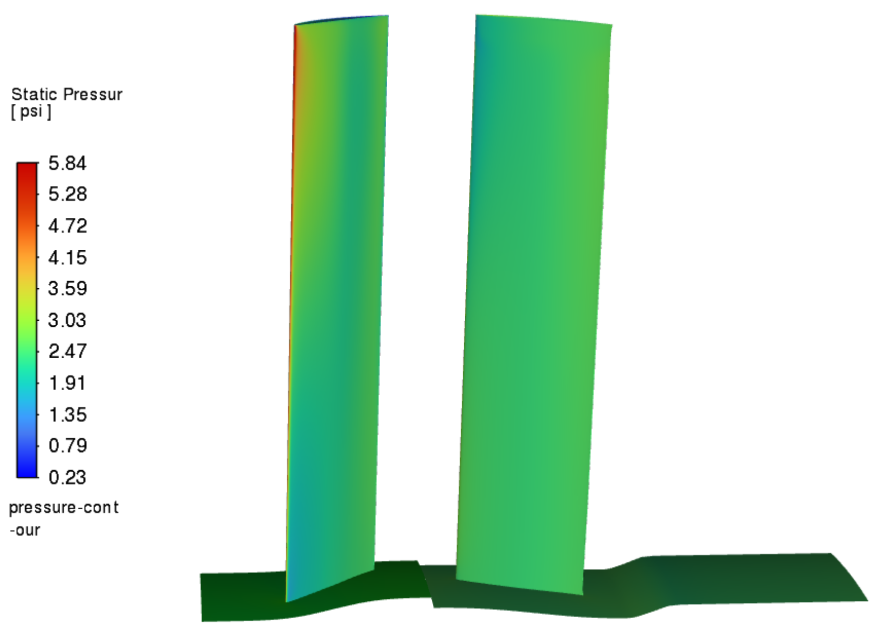


Figure 32: Static Pressure Contour Across Section Model of Stage 1

The static pressure on stage 1 rotor is the highest along the leading edge, which is expected from flow hitting an airfoil.

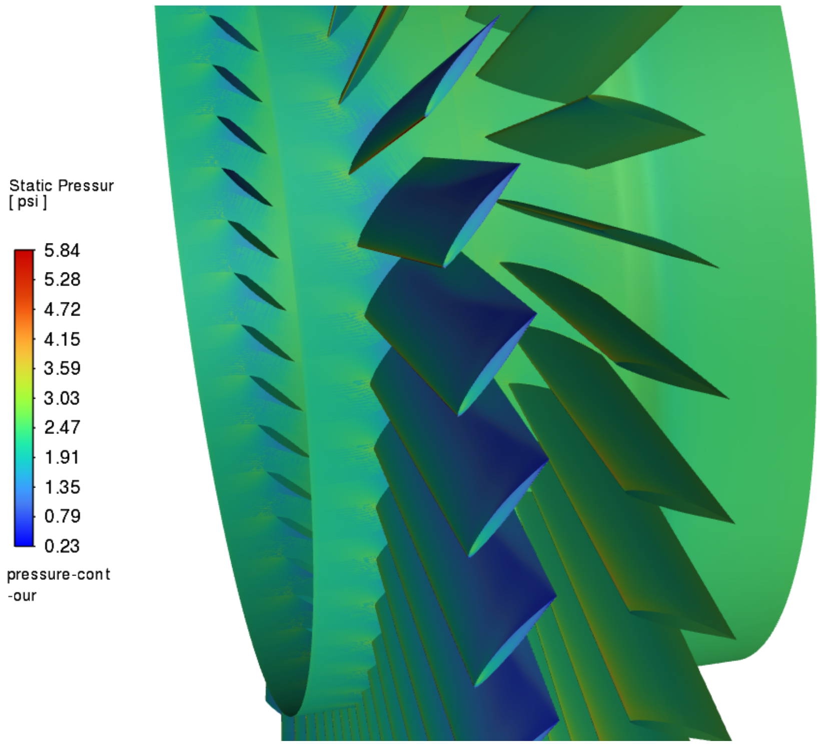


Figure 33: Zoomed Static Pressure Contour of Stage 1

The upper surface (suction surface) of the airfoil is a low-pressure zone because faster air speeds help drop the static pressure.

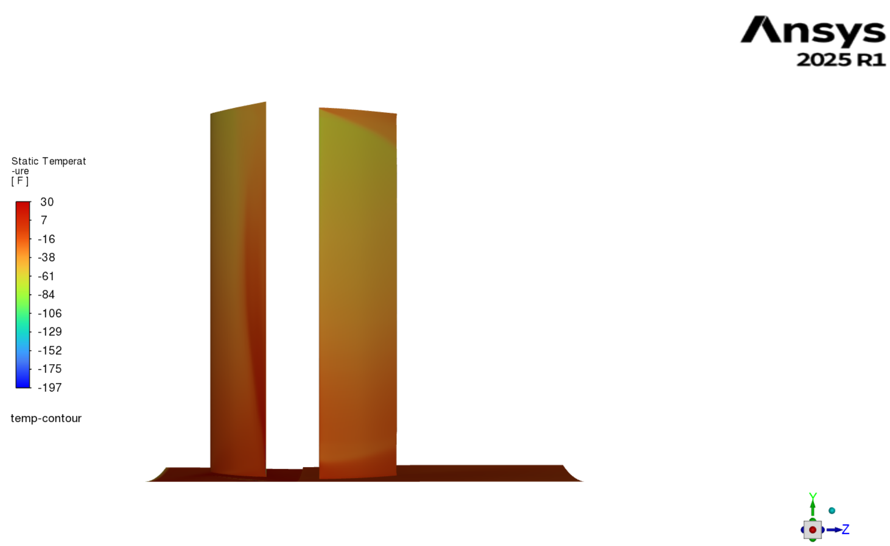


Figure 34: Static Temperature Contour Across Section Model of Stage 1

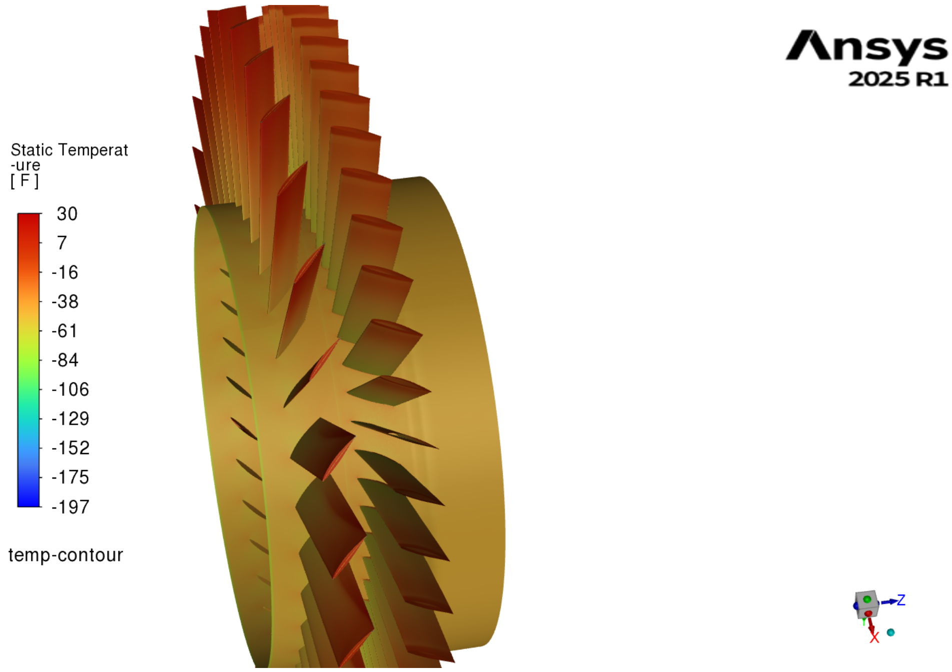


Figure 35: Static Temperature Contour of Stage 1

As the velocity over the top of the airfoil blade increases, to keep the total energy constant, the static temperature decreases.

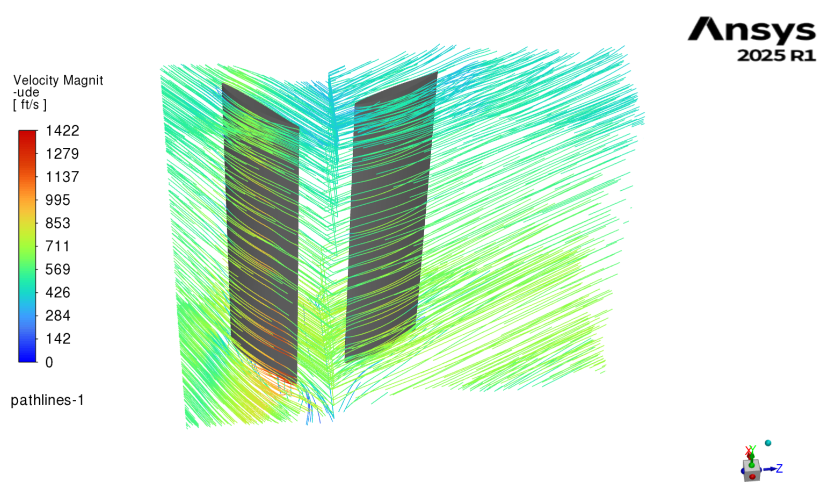


Figure 36: Velocity Streamline Across Section Model of Stage 1

The point where the rotor meets the shroud has a hot spot where there are higher velocities.

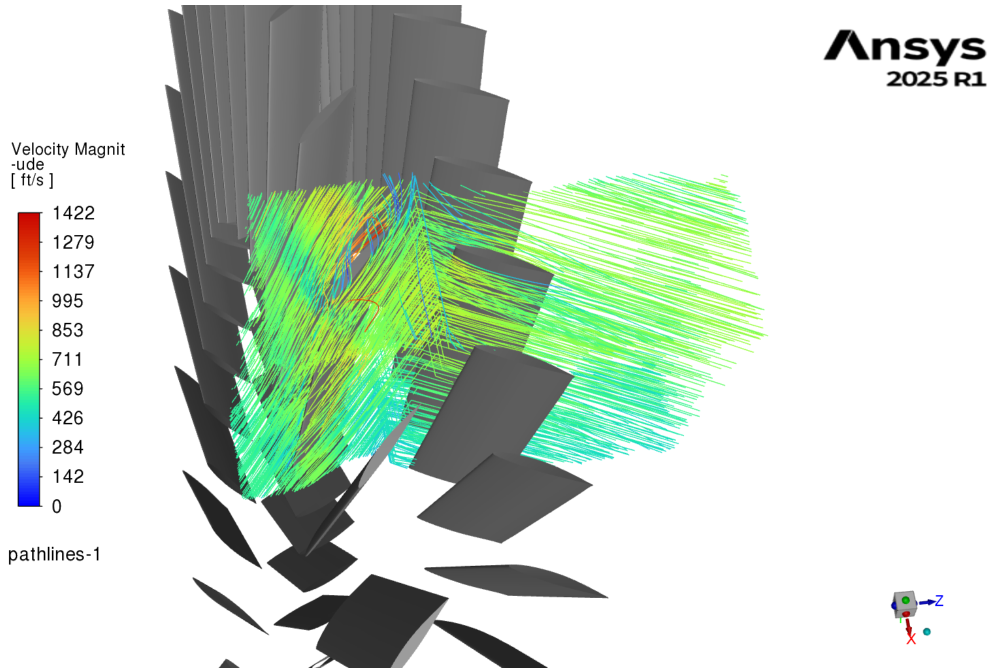


Figure 37: Velocity Streamline Across Stage 1

7.2.2 Stage 2

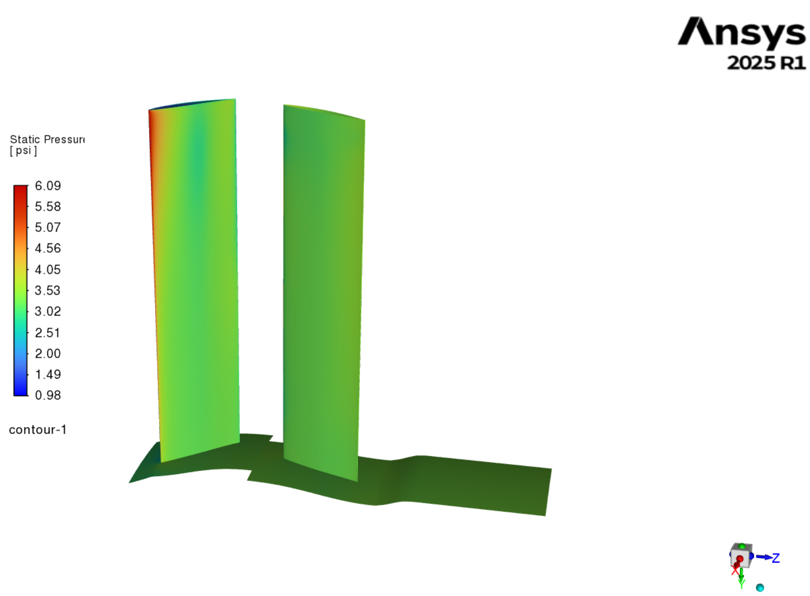


Figure 38: Static Pressure Contour Across Section Model of Stage 2

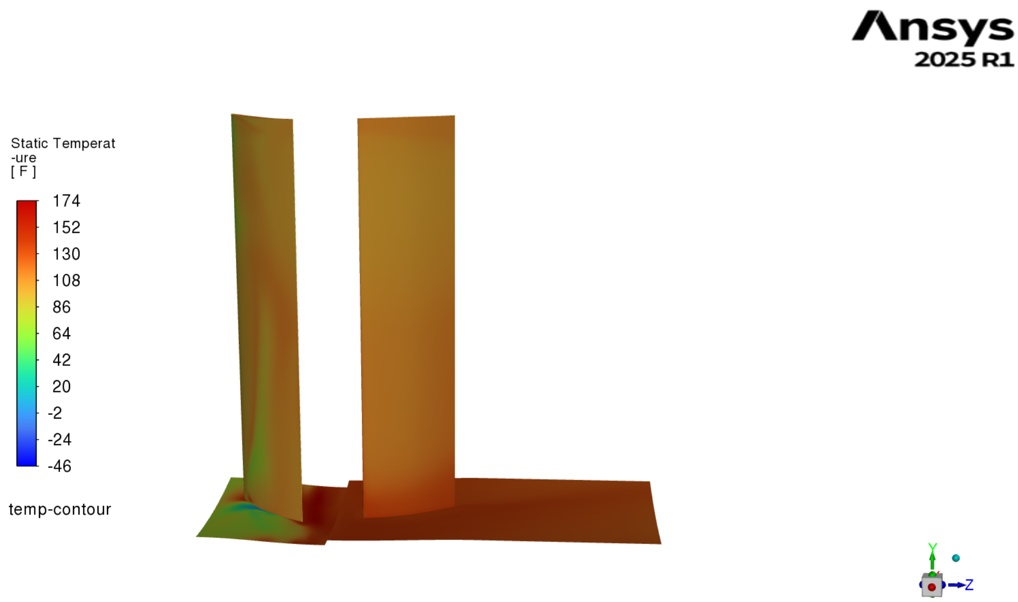


Figure 40: Static Temperature Contour Across Section Model of Stage 2

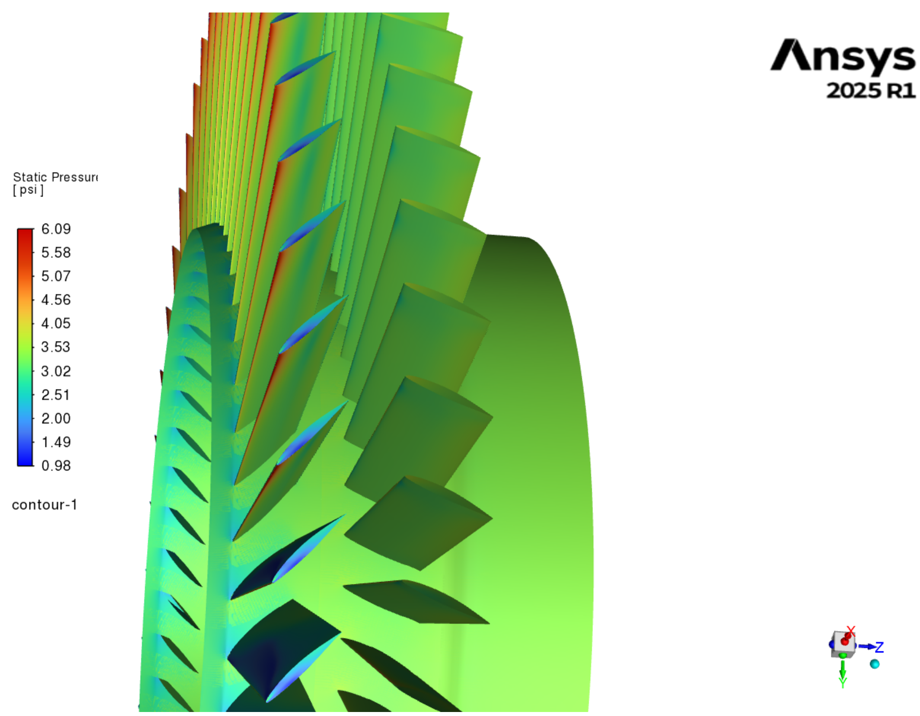


Figure 39: Static Pressure Contour Across Stage 2

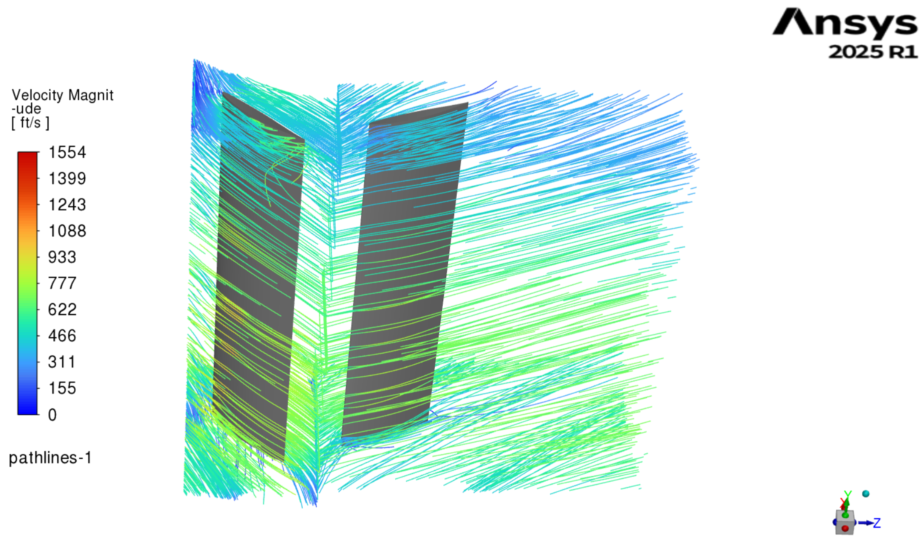


Figure 41: Velocity Streamlines Across Section Model of Stage 2

7.3 Interpolation Results at 40,000 ft

Using the two CPR values of 1.223 and 1.212 respectively, the rest of the compressor was interpolated through as prescribed in the two-point interpolation.

Table 11: Post Calc Tabulated CPR, P, and T Values @ 40,000 ft

Stage Number	**CPR**	P (Pa)	P (psi)	T (K)	T (°F)
1	1.223	18750	2.72	252	-5.7
2	1.212	22931	3.33	262	12.5
3	1.202	22725	3.30	267	21.5
4	1.193	22538	3.27	282	48.5
5	1.185	22368	3.24	298	77.3
⋮	⋮	⋮	⋮	⋮	⋮
20	1.112	20912	3.03	541	514.1
21	1.109	20856	3.02	559	546.5
22	1.106	20802	3.02	576	577.1
23	–	20751	3.01	593	607.7
Net CPR	22.061	–	–	–	–
Net Power Usage	229727 hp	–	–	–	–
Net Efficiency	11.5%	–	–	–	–

7.4 At sea level

At sea level, the pressure ratio of Stage 1 was 1.191 from the simulation.

7.4.1 Stage 1

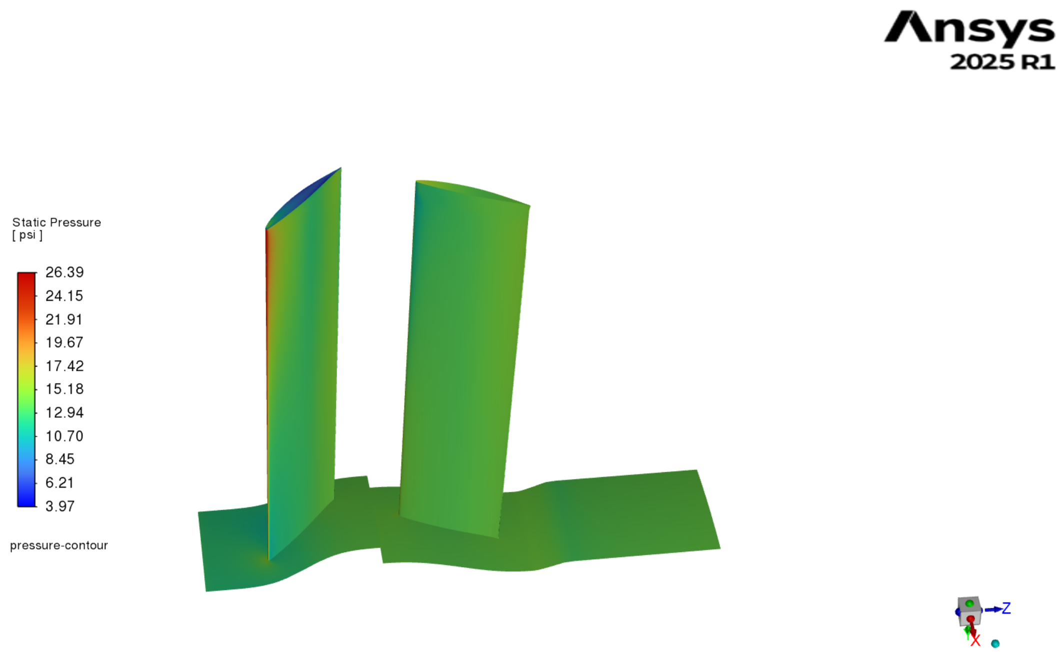


Figure 42: Pressure Contour Across Section of Stage 1

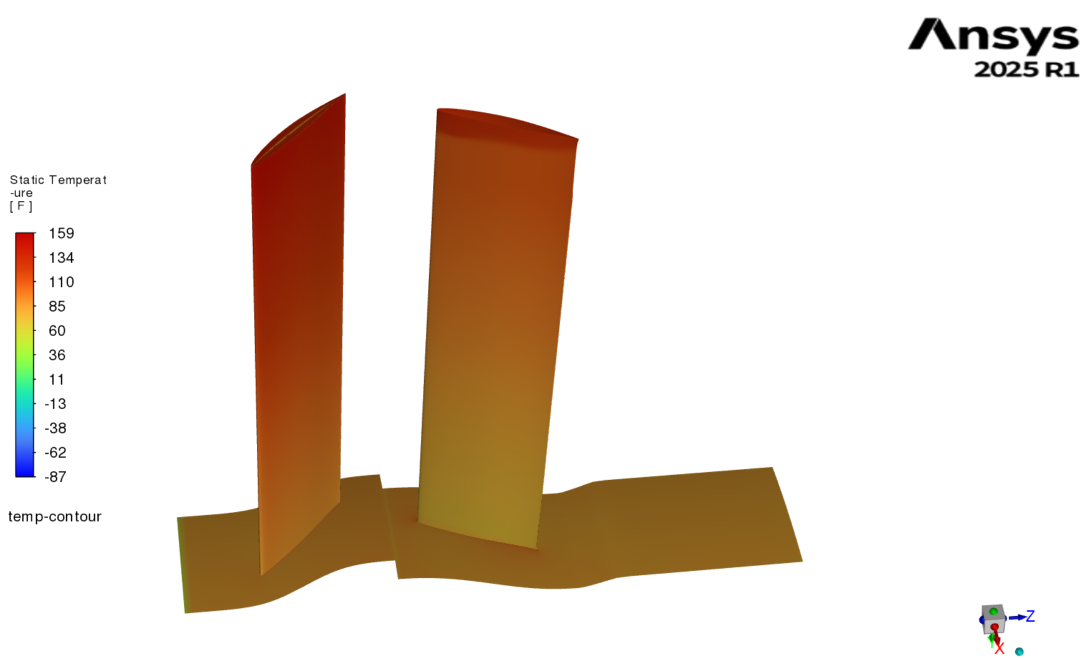


Figure 43: Temperature Contour Across Section of Stage 1

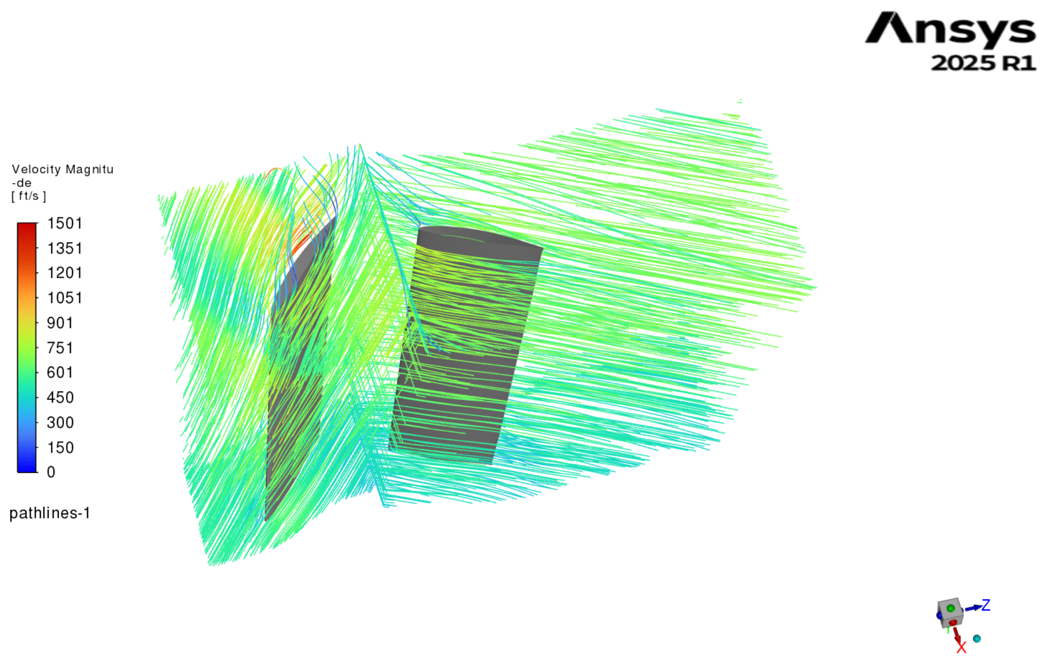


Figure 44: Velocity Streamlines Across Section of Stage 1

7.4.2 Stage 2

The Stage 2 pressure ratio was estimated by applying the same percentage change observed in Stage 1 between operating altitudes. This assumes that the relative performance scaling between stages remains consistent. Using the interpolation between CPR_1 and CPR_2 as prescribed, CPR_2 was determined to be 1.1775. This result will be used to interpolate throughout the rest of the compressor.

7.5 Interpolation Results at Sea Level

Table 12: Post Calc Tabulated CPR, P, and T Values @ Sea Level

Stage Number	**CPR**	P (Pa)	P (psi)	T (K)	T (°F)
1	1.191	101325	14.70	324	124
2	1.178	120678	17.50	333	140
3	1.166	119312	17.30	340	152
4	1.156	118126	17.13	357	183
5	1.146	117086	16.98	373	212
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
20	1.078	109500	15.88	588	599
21	1.076	109245	15.85	601	622
22	1.074	109007	15.81	614	646
23	–	108782	15.78	627	669
Net CPR	10.868	–	–	–	–
Net Power Usage	581671 hp	–	–	–	–
Net Efficiency	29.1%	–	–	–	–

7.6 Fully Tabulated Results

To make things easier for the reader, here are all of the tabulated results placed into one convenient location:

Table 13: Hand Calc Tabulated CPR, P, and T Values @ 40,000 ft

Stage Number	**CPR**	P (Pa)	P (psi)	T (K)	T (°F)
1	1.31	18750	2.72	252	-6
2	1.28	24500	3.55	272	30
3	1.25	31288	4.54	292	66
4	1.23	39180	5.68	311	101
5	1.21	48233	7.00	330	135
⋮	⋮	⋮	⋮	⋮	⋮
20	1.09	339471	49.22	577	579
21	1.08	368806	53.45	591	605
22	1.08	399153	57.92	604	628
23	–	430445	62.42	618	653
Net CPR	22.95	–	–	–	–
Net Power Usage	239424 hp	–	–	–	–
Net Efficiency	14.4%	–	–	–	–

Table 14: Hand Calc Tabulated CPR, P, and T Values @ Sea Level

Stage Number	**CPR**	P (Pa)	P (psi)	T (K)	T (°F)
1	1.17	101325	14.70	324	124
2	1.16	118534	17.19	339	151
3	1.15	137407	19.93	353	176
4	1.14	157969	22.91	368	203
5	1.13	180240	26.14	382	228
⋮	⋮	⋮	⋮	⋮	⋮
20	1.06	708576	102.76	564	556
21	1.06	754621	109.42	575	575
22	1.06	801602	116.27	585	593
23	–	849425	123.20	594	610
Net CPR	8.38	–	–	–	–
Net Power Usage	43506 hp	–	–	–	–
Net Efficiency	35.2%	–	–	–	–

Table 15: Post Calc Tabulated CPR, P, and T Values @ 40,000 ft

Stage Number	**CPR**	P (Pa)	P (psi)	T (K)	T (°F)
1	1.223	18750	2.72	252	-5.7
2	1.212	22931	3.33	262	12.5
3	1.202	22725	3.30	267	21.5
4	1.193	22538	3.27	282	48.5
5	1.185	22368	3.24	298	77.3
⋮	⋮	⋮	⋮	⋮	⋮
20	1.112	20912	3.03	541	514.1
21	1.109	20856	3.02	559	546.5
22	1.106	20802	3.02	576	577.1
23	–	20751	3.01	593	607.7
Net CPR	22.061	–	–	–	–
Net Power Usage	229727 hp	–	–	–	–
Net Efficiency	11.5%	–	–	–	–

Table 16: Post Calc Tabulated CPR, P, and T Values @ Sea Level

Stage Number	**CPR**	P (Pa)	P (psi)	T (K)	T (°F)
1	1.191	101325	14.70	324	124
2	1.178	120678	17.50	333	140
3	1.166	119312	17.30	340	152
4	1.156	118126	17.13	357	183
5	1.146	117086	16.98	373	212
⋮	⋮	⋮	⋮	⋮	⋮
20	1.078	109500	15.88	588	599
21	1.076	109245	15.85	601	622
22	1.074	109007	15.81	614	646
23	–	108782	15.78	627	669
Net CPR	10.868	–	–	–	–
Net Power Usage	581671 hp	–	–	–	–
Net Efficiency	29.1%	–	–	–	–

8 Discussion

The final compressor design met the main pressure ratio requirement at the design altitude. Based on the CFD results from the first two stages and the interpolation model, the 23-stage compressor produced a net pressure ratio of 22.061 at 40,000 ft. This satisfies the required minimum pressure ratio of 22:1. At sea level, the same design produced a lower net pressure ratio of 10.868. This is expected because the inlet air properties are different at sea level, and the same blade speed and geometry do not produce the same compression behavior at both operating conditions.

The CFD results show that each stage produces a modest pressure rise. At 40,000 ft, stage 1 produced a pressure ratio of 1.223 and stage 2 produced a pressure ratio of 1.212. Since the compressor uses many stages, these smaller pressure increases compound to produce the required total compression. This supports the use of a multistage axial compressor rather than attempting to achieve the full pressure ratio in a few stages, in line with the geometric limit on cross sectional area rather than length.

The temperature contours demonstrate the physically expected temperature increase through the compressor. Since work is added to the air by the rotor blades, the air temperature rises along with pressure. The temperature increase is an important design result because excessive exit temperature can reduce overall engine efficiency and may affect downstream components. The pressure and temperature contours generally follow expected compressor behavior, with higher pressure regions near blade leading edges and localized changes near the rotor and stator surfaces.

The streamline plots show that the flow is guided through the rotor-stator passages, but there are still regions of complex flow behavior near the blade surfaces and endwalls. These regions are expected in turbomachinery simulations because the air is being turned, accelerated, and diffused over a short distance. The stage 2 simulation is especially useful because it used the exported outlet profile from stage 1 instead of raw inlet air properties. This made the second stage result more representative of the actual staged compressor behavior. This was reflected when stage 2 stall was observed on its stator before its rotor, whereas stage 1 stalled at its rotor first.

One major limitation of the model was the meshing difficulty caused by the blade spacing. To obtain usable meshes, the blades had to be separated circumferentially more than originally desired. This reduced the number of rotor blades and stator vanes that could be used in each stage. The final model used 32 rotor blades and 32 stator vanes per stator, whereas previously over 50 blades were desired. A higher blade count would likely increase solidity and could improve flow turning, pressure recovery, and stage pressure ratio. Therefore, the simulated design may underpredict the performance of a more tightly packed blade row that could be achieved with improved meshing or a more refined geometry workflow.

Another limitation is that only the first two stages were directly simulated. The remaining stages were estimated using the interpolation model from the hand calculations. This was necessary because simulating all 23 stages directly would have required significantly more computational time. While the interpolation model gives a reasonable estimate of total compressor performance, it does not fully capture all downstream stage interactions, secondary flows, losses, or cumulative changes in blade incidence. The results should therefore be interpreted as a design-level estimate rather than a fully resolved final engine compressor simulation.

The efficiency values were also relatively low, especially at the maximum altitude condition. This indicates that although the design reaches the target pressure ratio, it does so with high power usage. Some of this is due to the simplified blade geometry, reduced blade count, and lack of detailed optimization. The compressor was designed primarily to meet the pressure ratio requirement, but further work would be needed to reduce power consumption and improve efficiency.

Future work should consider inlet guide vanes (IGVs). IGVs could be used to control the inlet flow angle entering the first rotor, reducing incidence losses and improving the pressure ratio of the first stage. In addition, blade twist should be included in future versions of the design. The current model uses a simplified blade geometry, but a real compressor blade should vary in angle from hub to tip because the blade speed changes with radius. Adding blade twist would allow the blade angle to better match the local relative flow direction, improving efficiency and reducing losses across the span of the blade.

Overall, CFD was useful for guiding the design because it provided pressure ratio, temperature, and streamline results that could not be obtained from hand calculations alone. The hand calculations established the expected compressor behavior and provided a way to extrapolate the full-stage performance, while CFD was used to evaluate actual stage behavior and identify the maximum stable pressure ratio before stall.

9 Conclusion

A 23-stage axial compressor was designed and analyzed for a gas turbofan engine. The compressor has a shell outer diameter of 66 in, an overall length of 118 in, and operates at 4000 rpm. Each stage uses a rotor-stator pair, with 32 rotor blades and 32 stator vanes per stage.

At the design altitude of 40,000 ft, the CFD-supported interpolation model predicted a net compressor pressure ratio of 22.061. This satisfies the required minimum pressure ratio of 22:1. The predicted power usage at this condition was 229,727 hp, with an estimated operating efficiency of 11.5%. At sea level, the compressor produced a predicted pressure ratio of 10.868, with a predicted power usage of 581,671 hp and an estimated operating efficiency of 29.1%.

The compressor was analyzed by simulating the first two rotor-stator stages in Ansys Fluent Turbo Workflow. The stage 1 outlet profile was exported and used as the inlet condition for stage 2, allowing the second stage to account for the flow field produced by the first stage. The remaining stages were estimated using the pressure ratio interpolation model developed in the hand calculations.

The final design meets the main pressure ratio requirement, but the model has several limitations. Mesh difficulty required the blades to be spaced farther apart circumferentially, reducing the number of blades that could be included in each stage. This likely reduced stage performance compared to a more optimized blade count. Additionally, the full compressor was not directly simulated, so the final pressure ratio and efficiency rely on extrapolation from the first two stages.

Future design work should focus on improving compressor efficiency. The most important improvements would be the addition of inlet guide vanes, optimized blade twist, and a higher blade count if meshing allows. These changes would improve flow alignment, reduce losses, and potentially increase the stage pressure ratio while reducing the power required to operate the compressor.

9.1 Design Time Estimate

In total, the design of the compressor took 200 hours, accounting for conceptual design, CAD modeling, meshing, CFD simulation, hand calculations, troubleshooting, and report preparation.

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10 Appendix

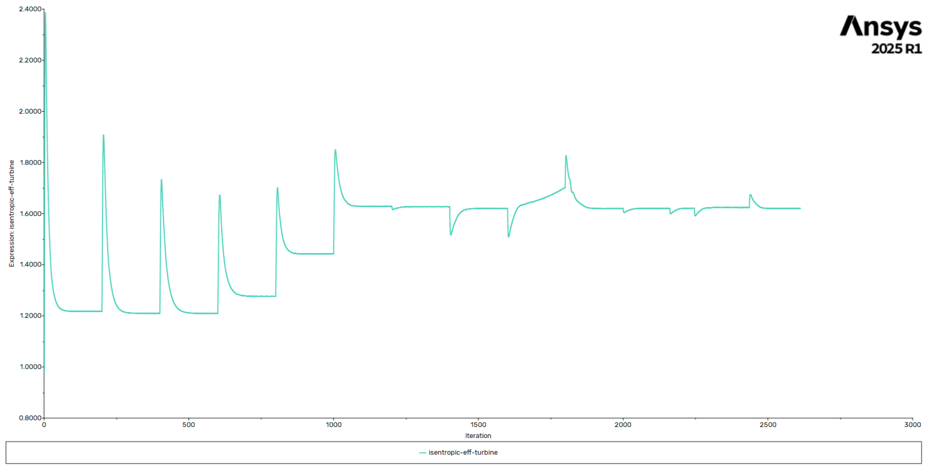

Figure 45: Stage 1 Isentropic Efficiency

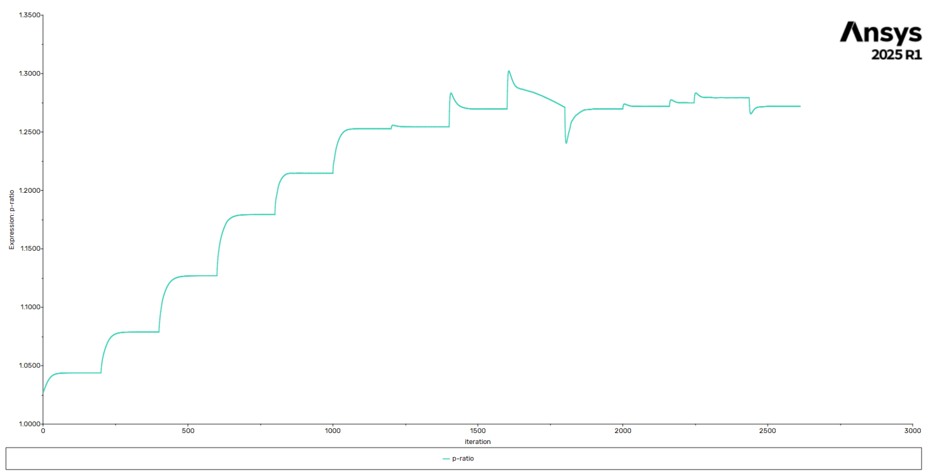

Figure 46: Stage 1 Pressure Ratio

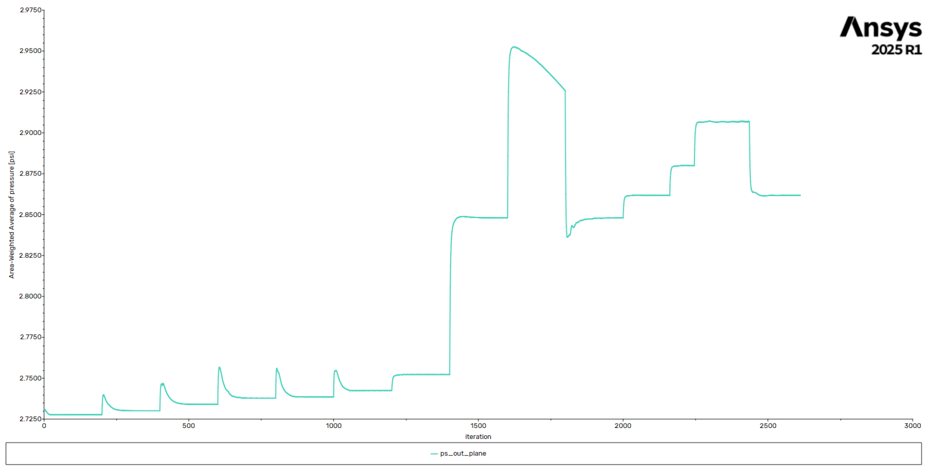

Figure 47: Stage 1 Pressure Outlet

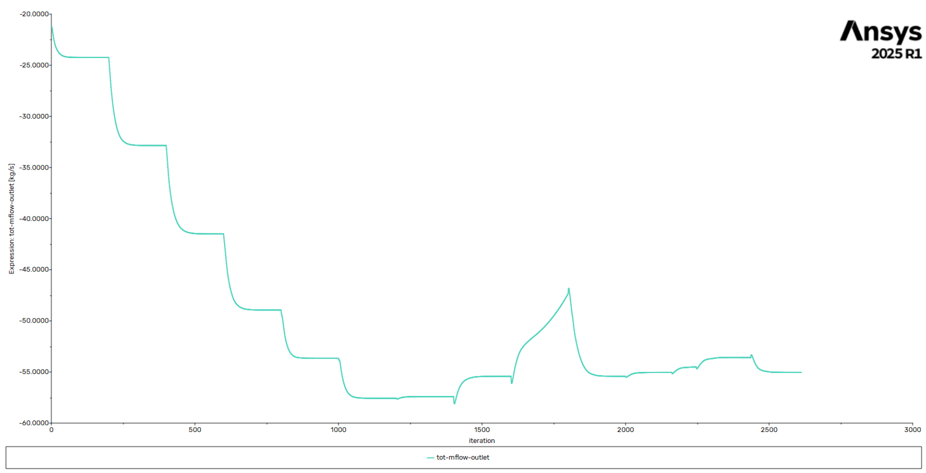

Figure 48: Stage 1 Total Outlet Mass Flow

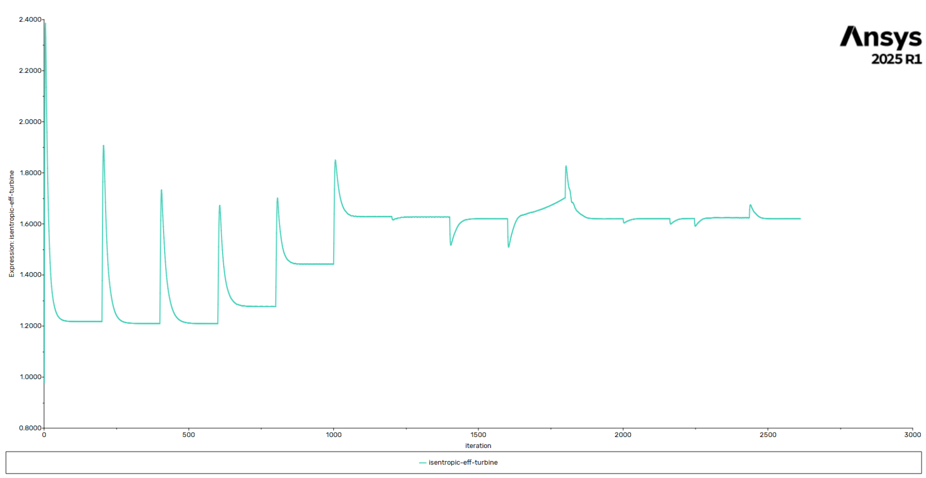

Figure 49: Stage 2 Isentropic Efficiency

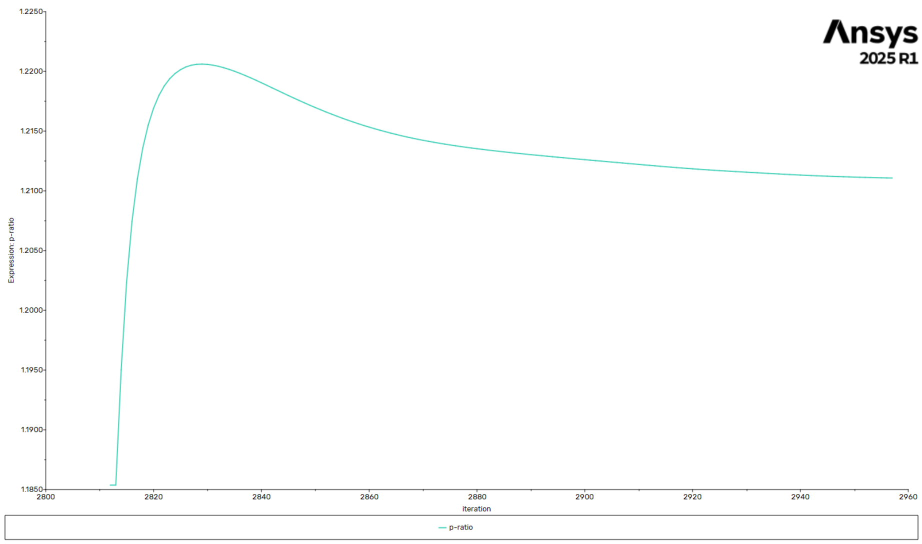

Figure 50: Stage 2 Pressure Ratio

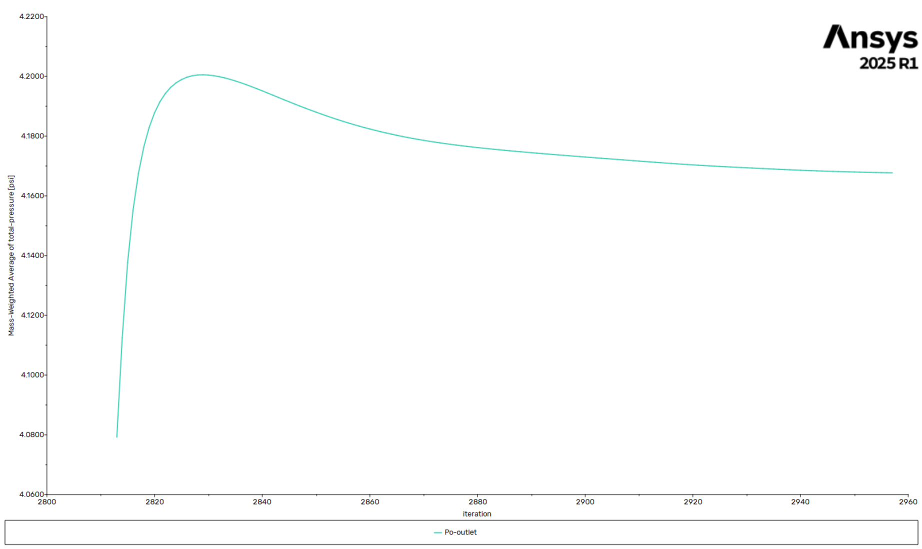

Figure 51: Stage 2 Pressure Outlet

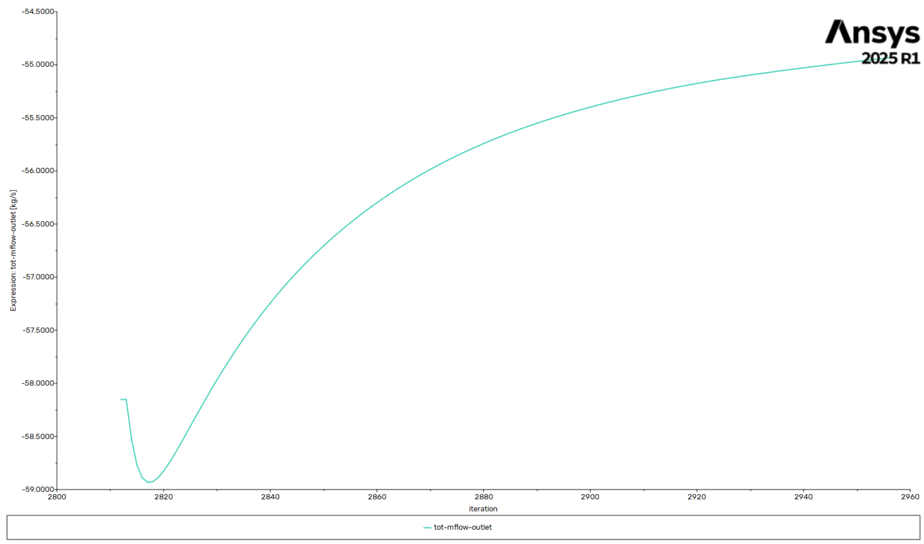

Figure 52: Stage 2 Total Outlet Mass Flow